

SAME CHOICE, DIFFERENT MIRRORS: INFORMATION COSTS ACROSS REPRESENTATIONS

PRELIMINARY DRAFT: PLEASE DO NOT CIRCULATE

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ABSTRACT. Rational inattention models treat information costs as structural features of decision makers, yet the same information choice can often be represented either as an experiment or as the posterior beliefs induced by that experiment. We design a laboratory experiment to test whether the information-cost structure revealed from choice is consistent across these theoretically equivalent representations. Subjects choose from the same feasible set of information levels either by selecting a posterior-accuracy level or by selecting the corresponding experiment-based sampling fraction. We vary both prior beliefs and the reward from a correct decision and evaluate behavior within each treatment using Prior-Invariant (PI) and Uniformly Posterior-Separable (UPS) cost benchmarks. Importantly, neither treatment is assigned to a particular cost class ex-ante. In a second section of the experiment, subjects also choose the representation under which to make an incentivized information decision, allowing us to examine whether the treatment differences persist when the representation is selected by the decision maker rather than externally assigned.

Information acquisition differs systematically across treatments even though the underlying states, incentives, and feasible information choices are held fixed. A substantial share of subjects exhibit choice patterns that are not fully consistent with either benchmark, and many subjects change the cost class with which their behavior is more consistent across treatments. Aggregate comparative statics similarly resemble different information-cost structures under the two representations. Because PI and UPS are theoretically distinct and generically disjoint under standard conditions, these changes cannot be dismissed as an innocuous reparameterization of a common cost function. The results demonstrate that theoretically equivalent representations of information need not be behaviorally equivalent and raise concerns about transporting revealed information costs across elicitation environments without first establishing the consistency of their representations.

Keywords: Rational inattention; costly information acquisition; revealed information costs; prior invariance; posterior separability; representation dependence; decision consistency

JEL Classification Numbers: D81, D83, C91

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Acknowledgments: This study is funded by the Decision Science Collaborative at Ohio State University. This study is pre-registered in AEA RCT Registry (AEARCTR-0015667): Kim, Suha, Hyeonggyun Ko and Jason Tayawa. 2026. “Costly Information Revealed by Beliefs: A More Direct Test of Rational Inattention Models.” AEA RCT Registry. April 19. <https://doi.org/10.1257/rct.15667-2.0>.

1. INTRODUCTION

Rational inattention models represent information acquisition as an optimal choice governed by its decision benefits and cognitive costs. However, the same information choice can often be represented in more than one way. A decision maker may choose a signal-generating technology, such as how intensively to search or sample, or may instead choose the accuracy of the beliefs that the technology produces. This distinction arises in many settings in which decision makers gather information about uncertain features of the environment before taking an action. When shopping online, a consumer may decide how many product reviews to read or how confident she wants to become that a product is of sufficiently high quality. A family choosing a school may decide how extensively to investigate academic outcomes, programs, and student experiences, or how certain it wants to be that a school is a good fit. Similarly, an employer may choose how many interviews, work samples, or references to collect, or how much confidence to require before concluding that a candidate is well matched to the position. These can be mathematically equivalent representations of the same information choice. An *experiment* induces a Bayes-plausible *distribution of posterior beliefs*, and an appropriate *distribution of posteriors* can be implemented by an *experiment*. This equivalence, however, is an equivalence from the analyst’s perspective. It does not imply that decision makers perceive the two representations as the same choice, or that they reveal the same cost of information under each representation.

This distinction matters because information costs are commonly treated as structural features of decision makers. A central objective of the rational-inattention literature is to infer the structure of information costs from observed behavior. [Caplin and Dean \(2015\)](#) provides a revealed-preference characterization of choice data that can be rationalized by optimal costly information acquisition and shows the extent to which the underlying information costs can be recovered. Subsequent works impose and test additional structure on those information costs. [Caplin et al. \(2022\)](#) provides behavioral characterizations of more structured cost classes, including posterior-separable and uniformly posterior-separable costs, and develops methods to identify these restrictions of the subclasses from choice data. [Denti \(2022\)](#) similarly distinguishes general responsiveness to information incentives from the more restrictive predictions

of commonly used specifications such as Shannon mutual information. These approaches typically recover or classify an information cost within a given representation of the information-acquisition problem. Together, these contributions make information costs empirically meaningful objects rather than merely convenient modeling assumptions, in that the resulting cost function can then be interpreted as a property of the decision maker and may be transported to other decision environments.

The existing identification results, however, are conditional on how the information-acquisition problem is represented to the decision maker. This qualification is potentially important because experiment-based and posterior-based costs are not generally interchangeable. [Denti et al. \(2022\)](#) distinguish costs assigned to the experiment performed from costs assigned to the induced distribution of posterior beliefs. They show that posterior-based costs may be inconsistent with a primitive model of costly experimentation, and that the distinction matters in equilibrium settings in which beliefs are endogenous. On the empirical side, [Dean and Neeligh \(2023\)](#) test whether subjects adjust attention in response to incentives and whether their choices satisfy the more restrictive predictions of Shannon mutual-information costs. They find support for incentive-responsive information acquisition but reject the linear Shannon specification in favor of a more flexible cost model. These studies establish that observed behavior can discriminate among models of information cost, but they do so within a given representation of the information-choice problem. They do not ask whether the cost structure recovered from the same individual remains stable when the same feasible information choices are presented in a theoretically equivalent representation. Yet mathematical equivalence between experiments and posterior beliefs does not guarantee this form of behavioral equivalence.

This paper examines this missing requirement. We ask whether the information-cost structure revealed by a rationally inattentive decision maker is invariant to the representation of the information choice itself. In our environment, each sampling experiment corresponds one-to-one with a posterior-accuracy level for each prior belief. We therefore present the same feasible information choices either as choices over posterior accuracy or as sampling decisions, while holding fixed the underlying states, payoffs, and information technologies. A standard formulation of rational inattention represents information by the distribution of posterior beliefs induced by a signal structure. [Caplin and Dean \(2015\)](#)'s revealed-preference framework follows this posterior distribution representation when characterizing costly information acquisition

from stochastic choice data. Our experiment uses this posterior-based tradition as one benchmark, and treats subjects’ intended posterior beliefs as the appropriate variable to measure. We compare the two representations by seeing whether the benchmark posterior-based choices extend to the experiment-based choices.

We then test whether the same individual reveals comparable information choices and thus the same information cost model rationalizability across the two representations. This is to structurally answer the question “Are the two representations of decision problems ‘practically’ equivalent?” The model classes of rational inattention that we are mostly interested in discussing the rationalizability are “Uniformly Separable Posterior (UPS)” and “Prior Invariant (PI)” classes. More interestingly, recent findings by [Denti et al. \(2022\)](#) and [Bloedel and Zhong \(2020\)](#) show that a UPS model cannot arise from an information cost structure based on costly experimentation, except when the cost of running experiments is nil. Since PI models are precisely those where costs stem from costly experimentation, this result implies that UPS and PI models are generically disjoint.¹ That is, no rational inattention model belongs to both classes. Rather than doing Afriat-type Revealed Preference testing for PI cost class, we define an environment that can clearly separate the implications between the two classes. Similar to the study of [Caplin and Dean \(2015\)](#), we could do a revealed-preference approach and characterize the behaviors of PI cost class in a more general environment. Our contribution is to design a useful restricted environment that can clearly separate the implications between the two classes, and actually document the distinct behaviors allowable by different classes of models, especially the PI class type behaviors that have not been frequently documented empirically. Our design identifies whether observed information choices are rationalizable by the restriction of a PI or UPS cost to the experimental domain. We do not nonparametrically identify the cost function outside this domain. While maintaining that the underlying cost belongs globally to either the PI or UPS class, the restricted environment provides diagnostic behavioral restrictions that can distinguish the two classes for some choice patterns.

One might argue that the underlying cost class could instead be identified through its behavioral restrictions, without directly manipulating the representation of the information

¹This does not assume that a posterior-based representation mechanically induces UPS-allowable behavior, or that an experiment-based representation induces PI-allowable behavior. Both posterior-based and experiment-based cost benchmarks can be evaluated in either elicitation frame. The empirical question is whether changing the representation of the same feasible information choices changes the revealed pattern of information acquisition.

choice itself. For UPS costs, one such route builds on sequential optimization. [Bloedel and Zhong \(2025\)](#) provide a complementary optimization foundation. They characterize Sequential Learning-Proofness (SLP) as the reduced-form restriction implied by flexible sequential cost minimization and show that, under an additional regularity condition, SLP costs are UPS. Their analysis also indicates that this criterion does not provide a symmetric way to compare the UPS and PI classes: commonly used PI specifications, such as likelihood-ratio costs, generally fail SLP, although prior invariance alone does not rule it out.

We therefore do not base our experimental comparison on a sequence of interim belief updates and endogenous stopping decisions. Instead, in the second section of the experiment, participants choose whether to make an incentivized information decision in the Posterior-based or the Experiment-based representation. This endogenous choice provides auxiliary evidence on which representation participants regard as more relevant to their decision process and allows us to examine whether the representation-dependent patterns documented in the first part persist when the elicitation frame is selected by the participant. At the same time, the design preserves a one-shot information-acquisition problem: participants attend to all components of a single signal and update their beliefs only once, limiting the additional complexity and fatigue that would arise from repeated updating and optimal-stopping decisions.

To the best of our knowledge, little research has examined which category individuals fall under when choosing the level of informativeness in a direct question where subjects explicitly choose their intended random posteriors, which follows the traditional convention of the decision variable in rational inattention models. This direct design approach would help our understanding of whether a random posterior distribution is simply a convenient analytical tool or a conceptually understandable object in practice. To achieve the direct approach, our design is carefully chosen to realize a simple decision environment that operationalizes the concept of posterior distributions, while observing the choice of information structure for us to control the objective information the subject receives. In addition, we also choose a design where subjects pay close attention to each unit of information based on their desired posterior distributions.

2. THEORIES OF RATIONAL INATTENTION AND MODEL CLASSES

This section develops the information-cost benchmarks used to organize our empirical analysis. Our objective is to determine whether information choices observed in our restricted

experimental environment are rationalizable by Prior-Invariant (PI) or Uniformly Posterior-Separable (UPS) costs. We first review the general costly-information-acquisition framework and define the two cost classes. We then specialize the framework to our *Black Swan* information technology, under which the same information choice admits a one-to-one experiment-based and posterior-based representation. Finally, we derive behavioral restrictions that allow us to distinguish the two cost classes on the experimental domain.

2.1. Models of Rational Inattention

Our framework belongs to the class of costly information acquisition models in which the cost of information is additively separable from the expected benefit of improved decision making, following [Caplin and Dean \(2015\)](#). We adopt the setup and notation of [Caplin and Dean \(2015\)](#) and [Dean and Neligh \(2023\)](#) to discuss the testable implications of rational inattention models.

A decision maker faces a finite state space Ω and a finite set of actions A . The payoff from choosing action $a \in A$ in state $\omega \in \Omega$ is denoted by $u(a, \omega)$. A decision problem is given by a full-support prior belief $p \in \Delta^\circ(\Omega)$ and an available action set (A), and the utility the decision maker gains from the action.² Before taking an action, the decision maker may acquire information about the state.

We represent an information structure as a stochastic mapping $\pi : \Omega \rightarrow \Delta(\Gamma)$, where Γ is the set of possible signals. Since each signal induces a posterior belief, we simply identify each signal $\gamma \in \Gamma$ with its associated posterior $\gamma \in \Delta(\Omega)$. Given prior p , each signal $\gamma \in \Gamma$ with positive unconditional probability satisfying $\pi(\gamma) \equiv \sum_{\omega \in \Omega} p(\omega) \pi(\gamma|\omega) > 0$ induces a posterior $\gamma \in \Delta(\Omega)$ according to Bayes' rule, $\gamma(\omega) = \Pr(\omega|\gamma) = \frac{p(\omega)\pi(\gamma|\omega)}{\sum_{\omega' \in \Omega} p(\omega')\pi(\gamma|\omega')}$. The experiment therefore induces a distribution Q_p^π over posterior beliefs satisfying Bayes' Plausibility: $\mathbb{E}_{Q_p^\pi}[\gamma] = p$, or $\sum_{\gamma \in \Gamma(\pi)} \pi(\gamma)\gamma = p$. This Bayes-plausibility condition allows the same information structure to be represented by its induced distribution of posterior beliefs.

Given a decision problem (p, A, u) , the decision maker chooses an information structure (π) to maximize expected payoff net of information costs:

$$G(p, A, \pi) - K(p, \pi).$$

²We restrict our attention to full-support priors. This avoids degenerate cases in which some states are assigned zero prior probability and hence cannot be reached by Bayesian updating.

The gross value of information $G(p, A, \pi)$ is the expected payoff obtained after observing the signal and choosing an optimal action conditional on the induced posterior. We can reformulate and express this gross value of information in the posterior-based formulation.³

We consider information cost functions K that are increasing in the informativeness of the induced posterior distribution and that normalize inattention to have zero cost. Let $\mathcal{K}$ denote the class of admissible attention cost functions. Two important subclasses are the uniformly posterior-separable class ($\mathcal{K}^{UPS}$) and the prior-invariant class ($\mathcal{K}^{PI}$), which are discussed in the later section for the classifications of optimal information choice behaviors.

2.2. Environment Restriction: Finding a “Black Swan”

We construct and introduce an environment that clearly indexes the information choices in either representation. Based on the specific information technology it has, we call it a “Black Swan” environment.

A “Black Swan” Environment:

- (1) We consider the binary state, binary action case: $\Omega = \{0, 1\}, A = \{0, 1\}$. We consider a decision maker who is willing to match the action to the state, $r \equiv u(0, 0) = u(1, 1) > u(a, \omega)$, with $a \neq \omega$ where $a, \omega \in \{0, 1\}$.
- (2) The information technology the decision maker faces consists of two types of signals, γ_0 and γ_1 , generated by the desired Q_p^π . One of the signals, the signal γ_1 , fully reveals the state, i.e., $\gamma_1(w = 0) \equiv \Pr(\omega = 0 | \gamma_1) = 0, \gamma_1(w = 1) \equiv \Pr(\omega = 1 | \gamma_1) = 1$.
- (3) From the fully revealing signal γ_1 in above description (2), if the true state is 0, then the possible signal that the decision maker will see is only γ_0 . If the true state is 1, then the signal γ_1 is realized with a probability, denoted by x . In short, the information structure is written as

$$\pi_x = \begin{matrix} & \gamma = \gamma_0 & \gamma = \gamma_1 \\ \begin{matrix} \omega = 0 \\ \omega = 1 \end{matrix} & \begin{pmatrix} 1 & 0 \\ 1 - x & x \end{pmatrix} \end{matrix}$$

³More formally, $G(p, A, \pi) = \sum_{\gamma \in \Gamma(\pi)} \pi(\gamma) [\max_{a \in A} \sum_{\omega \in \Omega} \gamma(\omega) u(a, \omega)]$, where $\pi(\gamma) = \sum_{\omega \in \Omega} p(\omega) \pi(\gamma | \omega)$. Reformulating and expressing this gross value of information in the posterior-based formulation, we write $Q \equiv Q_p^\pi$, and so we have $G(p, A, Q) = \sum_{\gamma \in \text{supp}(Q)} Q(\gamma) [\max_{a \in A} \sum_{\omega \in \Omega} \gamma(\omega) u(a, \omega)]$, subject to $\sum_{\gamma} Q(\gamma) \gamma = p$.

and the parameter x can uniquely index this information structure.

- (4) The decision maker chooses the informativeness (of experiment x or of the posterior γ_0) informative enough to choose the action 0 after seeing signal γ_0 , i.e., only allowing $x = 0$ and x such that $\gamma_0(0) \geq \frac{1}{2}$, or equivalently $x(p) \geq 1 - \frac{1-p}{p}$. The information is instrumental under this restriction.

Note that, even though the information technology is described through the information structure π_x and x that summarizes the information structure in the descriptions (2) and (3), we can always translate and map every experiment x to a posterior distribution Q_p^π (or we can write this as Q_p^x). For a given x , Bayes' rule implies $\gamma_0(\omega = 0) = \Pr(\omega = 0|\gamma_0) = \frac{1-p}{1-px}$,⁴ and the probability of observing the γ_0 signal is $\Pr(\gamma = \gamma_0) = 1 - px$. We also know that $\gamma_1(\omega = 0) = 0, \gamma_1(\omega = 1) = 1$. Thus, the random posterior Q_p^x for a given x can be equivalently written as

$$\begin{aligned} \gamma_0(\omega = 0) &= \frac{1-p}{1-px} \text{ with probability } 1-px, & \text{and} & \quad \gamma_1(\omega = 0) = 0 \text{ with probability } px. \\ \gamma_0(\omega = 1) &= \frac{p-px}{1-px} \text{ with probability } 1-px, & \text{and} & \quad \gamma_1(\omega = 1) = 1 \text{ with probability } px. \end{aligned}$$

Focusing on the state $\omega = 0$, this $\gamma_0(0)$ posterior belief can uniquely index the whole information technology, equivalently to the experiment x . When observing the signal γ_0 , the probability that the true state is 0 is the posterior belief $\gamma_0(0)$, and all the other entries of probabilities are determined as the decision maker chooses this probability $\gamma_0(0)$.

The description (4) enables us to restrict our interest to the instrumental (i.e., action-relevant) information levels chosen by the decision maker. That is, the decision maker chooses the information levels informative enough to take action 0 after seeing signal γ_0 , desiring the posterior belief at least as informative as it is more likely to do action 0. This equivalently restricts the range of experiment $x \geq 1 - \frac{1-p}{p}$, which is illustrated in the Figure 1. Notice that we consider the case where the decision maker always believes the true state is more likely to be state 1 ($p \geq 1/2$) to have this restriction meaningful. Since the equal prior $p = 1/2$ have this restriction exactly $x \geq 0$, the threshold $1 - \frac{1-p}{p}$ captures how close the prior p is to the equal prior, $p = 1/2$. When the prior p deviates more and more from the equal prior $1/2$, the experiment x has to be more and more restricted to greater fraction, meaning that the decision

⁴Here, we set p as the prior probability of the state $\omega = 1$.

maker should experiment for greater amount to do the action 1 as the prior belief goes closer to the state 1.

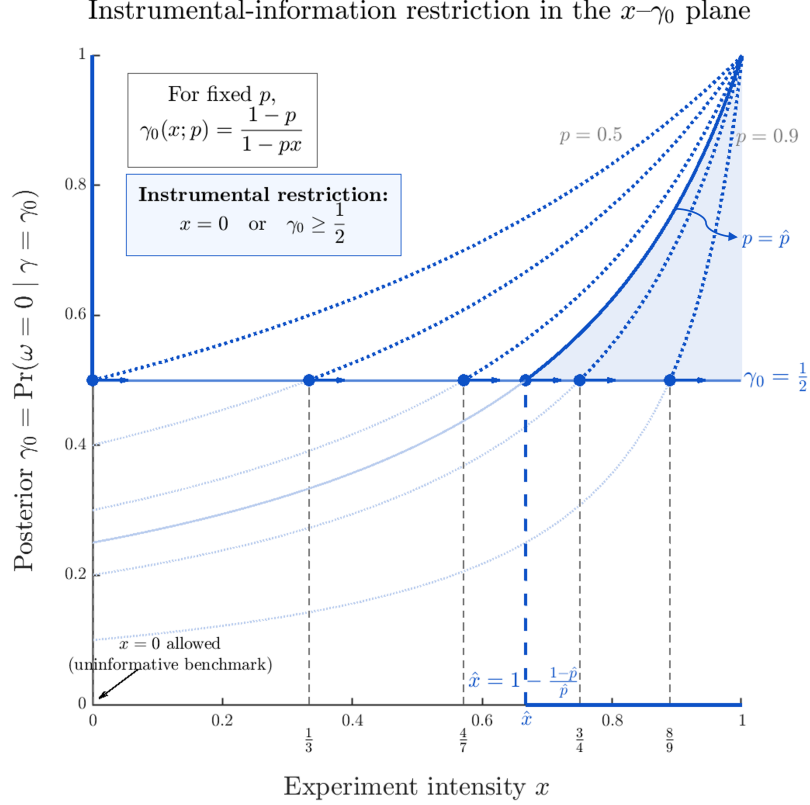


FIGURE 1. Instrumental Information Restriction

With this restriction, we interpret the experiment x as a “**sampling fraction** for the evidence of correct action (action 1)”, as it determines the chance of the signal γ_1 allowing a certain possibility of the false-positive diagnosis ($\frac{1-p}{p}$) despite the belief of the state being $\omega = 1$. Parallel to the interpretation, we interpret the posterior belief $\gamma_0(0)$ ($\gamma_0(1)$) as the “(in)**accuracy of action 0**” when observing the signal γ_0 , summarizing the decision maker’s optimal choice of informativeness with the information technology. (From now on, we will just write the posterior belief $\gamma_0 \equiv \gamma_0(0)$ for simplicity, although abusing the notation to some extent.)

2.3. Model Classifications and their Implications for the Posterior Elicitation

Variation in optimal information choices may reveal differences in the underlying structure of information costs. We therefore use the observed choice patterns to classify the information cost

structures that can rationalize behavior. To do so, we build on distinct classes of information cost functions developed in the rational inattention literature.

First, we present the classification of Uniformly Separable (UPS) Cost functions.

Definition 1. (*Uniformly Posterior Separable*)

A cost function is Uniformly Posterior-Separable (UPS) if there exists a strictly convex potential function $T : \Delta(\Omega) \rightarrow \mathbb{R} \cup \{+\infty\}$ such that, for every prior p and every Bayes-plausible distribution of posteriors $Q \in \Delta(\Delta(\Omega))$ with mean p ,

$$K(p, Q) = \sum_{\gamma \in \text{supp}(Q)} Q(\gamma) T(\gamma) - T(p).$$

Adapting to our Black Swan environment, this cost function can be rewritten simply as

$$K(p, \gamma_0) = \gamma_0 T(\gamma_0) - T(p).$$

We denote this class by $\mathcal{K}^{UPS}$:

$$\begin{aligned} \mathcal{K}^{UPS} := \{K \in \mathcal{K} \mid \text{There exists a strictly convex } T : \Delta(\Omega) \rightarrow \mathbb{R} \cup \{+\infty\} \\ \text{s.t. } K(p, \gamma_0) = \gamma_0 T(\gamma_0) - T(p)\}. \end{aligned}$$

We shall note that the same potential function T applies across priors. Thus, while the cost $K(p, \gamma_0)$ mechanically depends on the prior through the normalization term $T(p)$ and through the Bayes-plausible set of feasible posterior distributions, the underlying measure of posterior informativeness is invariant across decision problems. The Shannon mutual information cost is the canonical example, obtained by taking T to be negative Shannon entropy, up to a positive multiplicative constant.

Second, a cost is Prior Invariant (PI) if the same Blackwell experiment carries the same information cost regardless of the prior to which it is applied.

Definition 2. (*Prior Invariant*)

For all experiment $\pi \in \Pi$, where Π denotes the set of feasible information structures, and for all full-support priors $p, p' \in \Delta^\circ(\Omega)$,

$$K(p, Q_p^\pi) = K(p', Q_{p'}^\pi),$$

Notice that the posterior distributions Q_p^π and $Q_{p'}^\pi$ generally differ even though the underlying experiment π is held fixed.

Again adapting to our Black Swan environment, this cost function can be rewritten simply as

$$K(p, Q_p^x) = K(p', Q_{p'}^x), \text{ or } K(p, x) = K(p', x)$$

as the experiment $\pi \in \Pi$ can be uniquely represented through x in the environment.⁵

We refer to this cost class as $\mathcal{K}^{PI}$:

$$\mathcal{K}^{PI} := \{K \in \mathcal{K} \mid \text{For all experiment indexed by } x \in [0, 1] \text{ and}$$

$$\text{for all full-support priors } p, p' \in \Delta^\circ(\Omega), K(p, x) = K(p', x)\}.$$

This class captures the idea that the cognitive or technological cost of implementing a given Blackwell experiment is independent of the decision maker's prior belief. It is conceptually distinct from Uniformly Posterior Separability: UPS imposes a common posterior potential (T) across priors, whereas Prior Invariance imposes that the same signal-generating technology has the same cost across priors.

We now derive behavioral implications of PI and UPS costs within the Black-Swan environment. The restriction to a one-dimensional family of information structures allows both the experiment intensity x and the posterior γ_0 to index the same information choice. This makes it possible to express the restrictions imposed by the two cost classes in directly observable comparative statics.

Throughout this subsection, we focus on interior optimal choices in the action-relevant (i.e., information being instrumental) region. We consider the implication of Prior Invariant (PI) costs, distinguished disjointly from Uniformly Posterior Separable (UPS) costs, which are the form of the information cost functions broadly employed in the literature. A central behavioral implication of the UPS class is *Locally Invariant Posteriors*, studied by [Caplin et al. \(2022\)](#): holding payoffs fixed, if a set of posteriors is optimal for one prior, then the same posteriors remain optimal for nearby priors as long as the prior remains in the convex hull of those posteriors. We apply this behavioral implication in our environment. To do so, in addition to

⁵We can also write $K(p, \gamma_0)$ by indexing Q_p^x with γ_0 , which is equivalently induced by the experiment x , $\gamma_0(x; p) = \frac{1-p}{1-px}$.

the environment restrictions, we additionally impose a shape restriction on the cost function K .

Assumption 1. (*Sufficient Convexity of the cost function K*)

The information cost $K(p, x)$ is twice continuously differentiable on the relevant domain, and sufficiently convex locally at every interior optimal experiment $x^(p, r)$, such that $\frac{\partial^2}{\partial x^2} K(p, x^*) > \frac{rp(1-p)}{1-x^*(r, p)}$.*

This assumption is not part of the definition of either PI or UPS. Rather, it is an additional shape restriction that sharpens the comparative static predictions of PI costs in our restricted environment. In particular, it ensures that, under PI, the optimal experiment x^* responds sufficiently little to changes in the prior p for the effect on the experiment-to-posterior mapping to dominate.

Proposition 1. (*PI-Rationalizable Choices*)

Suppose the cost function K satisfies the Assumption 1. The cost function K satisfies that the marginal cost in experiment x does not depend on the prior p at the interior optimum, i.e.,

$$\frac{\partial^2}{\partial p \partial x} K(p, x^*) = 0$$

(This can be verified by the optimal experiment choices x^ satisfying $p \frac{\partial}{\partial p} x^*(r, p) = r \frac{\partial}{\partial r} x^*(r, p)$) if and only if the cost function K is rationalizable on the restricted domain by a strictly convex PI cost satisfying the sufficient convexity condition in Assumption 1.*

Moreover, this implies that the optimal posterior γ_0^ shows*

$$\frac{\partial}{\partial p} \gamma_0^* < 0,$$

Proof. The proof is included in the Appendix; see Appendix B.3. ■

This indicates that, for the cost K to be PI under the restricted environment with strong convexity, not only the cost function K itself but also the marginal cost in experiments (K_x) must not depend on the prior p . This also implies that with PI cost, the variation of the marginal cost in experiments, K_x , should be the same in the directions of the two parameters p and r , i.e., $px_p^* = rx_r^*$. As the marginal cost in experiments is $K_x(p, x^*) = rp$ at the optimum and so $x_p^* = \frac{r}{K_{xx}}$, $x_r^* = \frac{r-K_{xp}}{K_{xx}}$, we obtain $K_{xp} = 0$ if and only if $px_p^* = rx_r^*$.

We also consider the indication of Uniformly Posterior-Separable costs under our restriction. We apply *Locally Invariant Posteriors* in [Caplin and Dean \(2015\)](#) and present a version of it under our restriction. In our binary-state environment, this implies that the posterior following the signal γ_0 is locally invariant to changes in the prior.

Proposition 2. (*UPS-Rationalizable*)

Suppose the cost function K satisfies the Assumption 1. An optimal posterior choice schedule satisfying

$$\frac{\partial}{\partial p} \gamma_0^*(p, r) = 0$$

with a strict reward responsiveness $0 < \frac{\partial}{\partial r} \gamma_0^(p, r)$ is rationalizable on the restricted domain by a strictly convex UPS cost satisfying the sufficient convexity condition in Assumption 1.*

Moreover, in this case, $K_{xp}(p, x^) < 0$.*

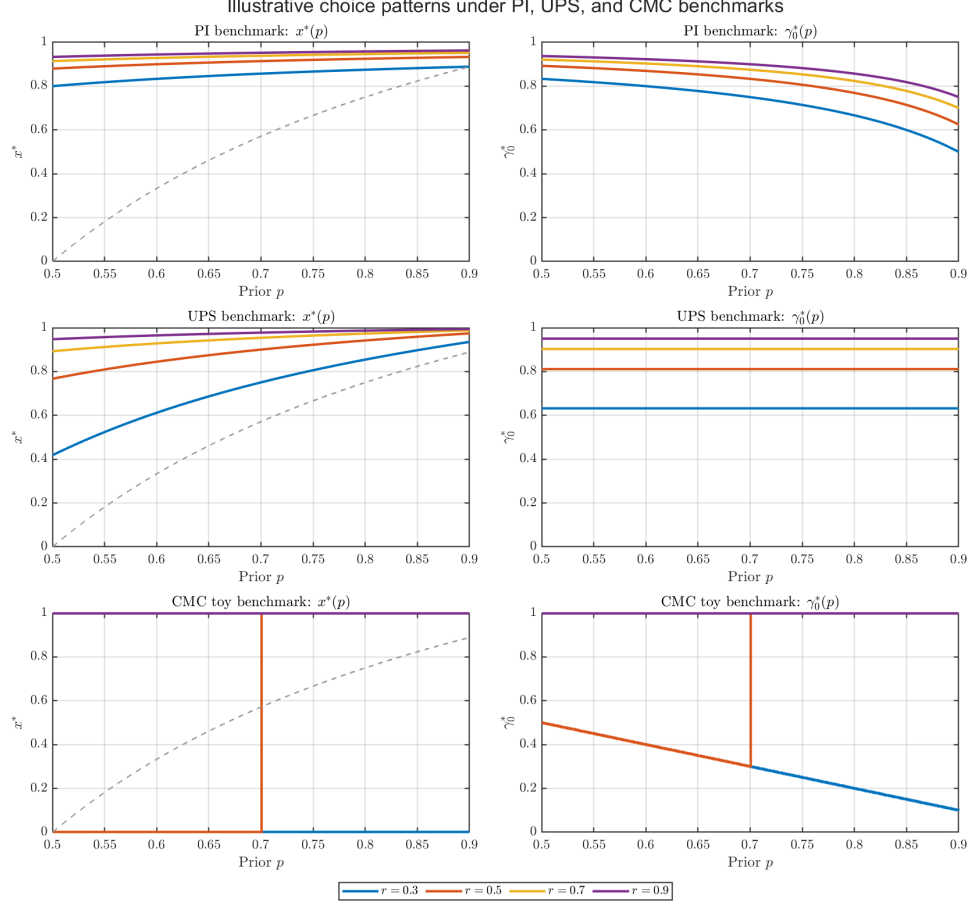
Proof. The proof is included in the Appendix. ■

This *Locally Invariant Posteriors* implication holds true within a general environment outside of our restriction, as studied in [Caplin and Dean \(2015\)](#).

In the Appendix, we additionally note that this further implies that when the cost is PI, the optimal experiment choice x^* is relatively stable in varying prior p , whereas when the cost is UPS, the optimal posterior choice γ_0^* is relatively stable in p . Therefore, we design and create a useful restricted environment that can effectively observe distinct behavioral patterns between the two classes.

In our *Black Swan* environment, depending on whether the decision maker's cost function K lies within $\mathcal{K}^{UPS}$ or within $\mathcal{K}^{PI}$, the decision maker's information choices (in posterior belief) $\gamma_0(0)$ prediction is vastly different in its pattern, presenting constant and decreasing choices of posterior belief (γ_0), respectively. Noticing these distinct patterns of behavior, our environment construction choice allows us to additionally identify and analyze the behavioral differences in terms of the models of cost functions between the ways of information representations.

For the analysis, we first classify individual information choice based on whether (i) γ_0 is constant in the prior, or (ii) γ_0 is decreasing in the prior. Then we further verify whether those cases satisfy (i)-a) $K_{xp}(p, x) < 0$ or (ii)-a) $K_{xp}(p, x) = 0$, respectively. From the propositions, we classify that (i)-a) indicates PI costs, and (ii)-a) indicates UPS costs. The analysis plan is



* Note: PI Benchmarks are on the first row, and the UPS Benchmarks are on the second row. The last row shows an example of a Constant Marginal Cost function. For the PI Benchmarks, the cost function $K(x) = -\kappa_{PI} \log(1 - x)$ is used. For the UPS Benchmarks, $\gamma_{UPS}^*(p, r) = 1 - \exp(-r/\kappa_{UPS})$, or equivalently, $x_{UPS}^*(p, r) = \frac{\gamma_{UPS}^*(p, r) + p - 1}{p \gamma_{UPS}^*(p, r)}$. For the Constant Marginal Cost case, we use $K(x) = \kappa_{CMC} x$.

FIGURE 2. PI and UPS Cost functions and the Optimal Information Choices

preregistered in the AEA-RCT registry, available at <https://doi.org/10.1257/rct.15667-2.0>.

3. EXPERIMENTAL DESIGN

We design an experiment that creates a simple and tightly controlled environment in which the relevant objects of the model can be observed. To effectively answer each relevant question, we divided the experiment into two sections. Experiment Section 1 first documents the patterns of the informativeness choices depending on the different information representations. It is

designed to answer first whether the informational choices under the posterior-based representation can be extended to the choices under the experiment-based representations. Experiment Section 2 then follows with the design to answer whether a specific information representation is employed by an individual when acquiring information. This is to track the underlying process of an individual who desires to acquire some information for later action. In other words, individuals may have a certain method to measure granular, unit information, and by observing the method, we may be able to pave a way to derive a welfare implication.

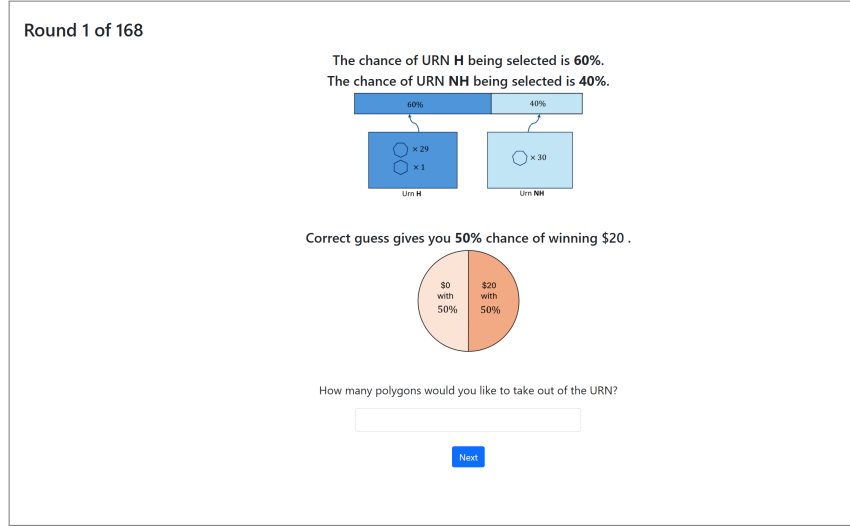
3.1. *Experiment 1: Documenting the Choice of Informativeness*

In all the experiments including the Experiment 1, subjects repeatedly face the simple task of guessing between two urns: Hexagon (H) and No-Hexagon (NH) Urn. Thus, there are two possible states, $\Omega = \{H, NH\}$, where H denotes the state in which the urn contains hexagons and NH denotes the state in which it does not. The subject chooses between two actions: guessing that the urn is H (denoted by h), or guessing NH (denoted by nh). Each urn has 30 polygons. In the No-Hexagon Urn, all 30 polygons are pentagons, whereas the Hexagon Urn consists of one hexagon and 29 pentagons.

When implementing the binary states and information technology, we use polygons rather than more immediately distinguishable cues, such as colored balls, so that processing each observation requires visual attention. Distinguishing polygons by the number of their sides requires subjects to inspect the observations rather than relying on an immediately salient feature such as color, while the perceptual similarity of polygons also provides an additional way to extrapolate the difficulty of distinction. For example, polygons with larger numbers of sides may be more difficult for participants to distinguish than geometrically more distinct shapes. This feature allows the information technology to remain objectively simple while making the processing of each observation attention-requiring. Related geometric perception tasks have been employed in the literature; in particular, [Caplin et al. \(2020\)](#) use regular polygons with 7 to 10 sides in an incentivized attention task.

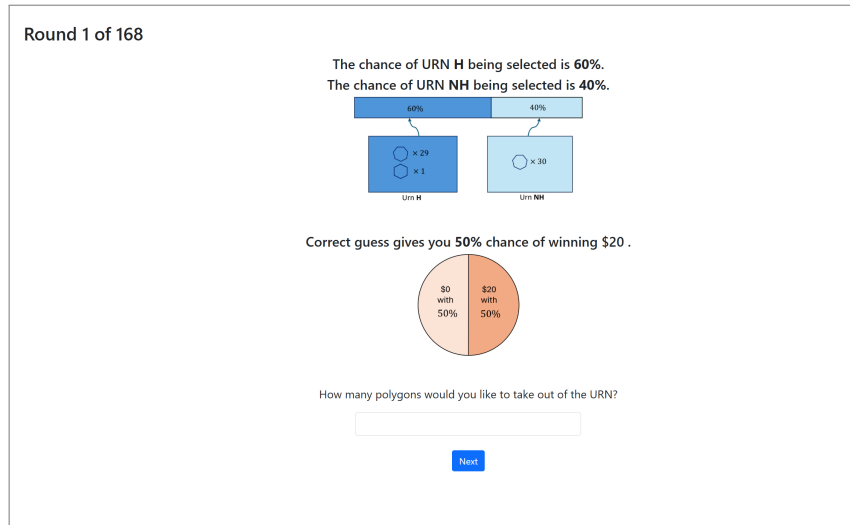
At the beginning of each round, the computer will randomly select one of two urns with probability $p = \Pr(H)$ (prior) that the Hexagon Urn is selected. The subject receives a fixed prize of \$20 with probability r if the guess is correct and receives zero otherwise: $u(h, H) = u(nh, NH) = 20r$, $u(nh, H) = u(h, NH) = 0$. We vary both the prior $p \in \{0.50, 0.55, 0.60, 0.65\}$

and the incentive probability $r \in \{0.30, 0.50, 0.70, 0.90\}$. The probabilistic prize structure allows us to vary the expected return to correct decisions while holding the prize amount fixed. All this information is presented to the subject on a single page, ensuring we do not incur additional costs from movement, such as scrolling or eye-gaze control, as shown in the examples in Figures 3 and 4.



How many polygons would you like to take out of the URN?

FIGURE 3. Example Screen of Treatment 2: Choosing Sampling fraction



How likely do you want the guess NH to be correct in case no hexagon appears?

FIGURE 4. Example Screen of Treatment 1: Choosing Posterior

Before making the final state guess, subjects can inspect the urn. They do so by taking out polygons from the assigned urn *without replacement*. For each draw, a polygon appears for

0.7 seconds, which then disappears, followed by another polygon.⁶ The number of polygons drawn is determined by the subject’s chosen level of informativeness. The representation of this elicitation varies by treatment. In the Posterior-based frame treatment, subjects choose an object that corresponds to their posterior belief. In the Sampling-based frame treatment, subjects choose an object corresponding to an information structure. After this choice, the subject observes a signal realized according to the selected informativeness. Notice that we do not restrict subjects from choosing all 30 polygons taken out for all rounds; subjects are guided to pay attention to each and every polygon that they take out during the experiment, and it is their choice to invest effort and time in all 30 polygons. There is no additional loss beyond the participant’s attention and patience in choosing 30 polygons, ensuring that the cost is solely the participant’s attention and its byproduct.

Here, the signal takes one of two forms: a sequence with no hexagon or a sequence with a hexagon. For notational simplicity and to be consistent with the description in the theoretical background, define $\gamma_0 =$ (sequence with no hexagon), $\gamma_1 =$ (sequence with a hexagon). In the Experiment-based frame treatment presented in Figure 3, the subject chooses how many polygons ($n \in \{0, \dots, 30\}$) to draw from the urn. In particular, we ask “*How many polygons would you like to take out of the Urn?*” The induced information structure is

$$\pi(n) = \begin{matrix} & \gamma = \gamma_0 & \gamma = \gamma_1 \\ \begin{matrix} \omega = NH \\ \omega = H \end{matrix} & \begin{pmatrix} 1 & 0 \\ 1 - \frac{n}{30} & \frac{n}{30} \end{pmatrix} \end{matrix}.$$

Thus, if the true state is NH , no hexagon can appear. If the true state is H , a hexagon appears with probability $n/30$, which is the probability x in our theoretical benchmark. Larger values of n correspond to more informative experiments.

In the Posterior-based frame treatment presented in Figure 4, we ask the subjects “*How likely do you want the guess ‘No Hexagon’ to be correct in case no hexagon appears?*” We elicit the posterior associated with the no-hexagon signal by using this question, $\gamma_0(NH) \equiv \Pr(NH|\gamma_0)$.

⁶The lasting time, 0.7 seconds, of the shapes is determined from the pilot results. In fact, this design choice is broadly consistent with evidence that abstract shape representations form within the time range of 0.7 seconds. For instance, in Baker and Kellman (2018) approximately 0.1-0.11 seconds is provided, and the strongest attentional interference between successive visual targets generally occurs within the first 0.2-0.5 seconds (Dux and Marois, 2009). Research on fixation-based and diffusion models of choice shows that visual exposure time governs the amount of evidence processed and accumulated during a decision (Armel et al., 2008; Krajovich et al., 2010, 2012).

Given the design, this posterior summarizes the chosen random posterior. Notice that when the subject ‘inspects the urn, there are two possible cases: either (1) they observe a hexagon, or (2) they don’t observe a hexagon. In the case that **(1) they observe a hexagon**, the subject is 100% certain that the assigned urn is the Hexagon Urn; that is, the posterior assigns probability one to state H , so that $\gamma_1(NH) = 0, \gamma_1(H) = 1$. In the other case that **(2) they do not observe a hexagon**, the subject is *not* 100% certain that the assigned urn is No Hexagon since both urns contain mostly pentagons. The question that is employed in the treatment implements this uncertainty of the subject and their probability (x) perceived for this uncertainty. Thus, their answer for the level of certainty fully reveals their intended posterior distribution. If a hexagon appears, the posterior assigns probability one to state H , so that $\gamma_1(NH) = 0, \gamma_1(H) = 1$. For a chosen n , Bayes’ rule implies $\gamma_0(NH) = \frac{1-p}{1-\frac{np}{30}} = \frac{30-30p}{30-np}$, and the probability of observing the no-hexagon signal is $\Pr(\gamma_0) = 1 - \frac{np}{30}$. The chosen random posterior can be written as

$$\gamma_0(NH) = \frac{30-30p}{30-np} \text{ with probability } 1 - \frac{np}{30}, \quad \text{and} \quad \gamma_1(NH) = 0 \text{ with probability } \frac{np}{30}.$$

This posterior representation is equivalent to the information-structure representation, as discussed in the previous section. Hence, under Bayesian updating, the Natural and Direct frames identify the same level of informativeness. Once the subjects report their intended random posterior, we translate their chosen level of “accuracy” to the appropriate number of polygons to be drawn to achieve that level. Notice that increasing the level of certainty means increasing the number of polygons to inspect from the assigned urn. By choosing 100% certainty, the subject elects to inspect all polygons from the assigned urn. By this design choice, the comparison across the two treatments allows us to test whether subjects respond equivalently when the same information acquisition problem is framed as choosing an experiment versus choosing an intended posterior.⁷ The timing of the decision-making during each treatment is summarized in Figure 5.

For the robustness, we conduct the third treatment session, which provides an exact Bayesian mapping between the number of draws n and the random posterior via a slider when subjects decide on n , whereas the second session does not. Each subject participates in three separate

⁷All the aforementioned information is clearly provided to subjects. The instructions of the experiments can be found in the Supplementary Materials, [Appendix A](#).

sessions, each consisting of 168 rounds of urn-guessing tasks. Two rounds are randomly chosen for payment at the end of each session.

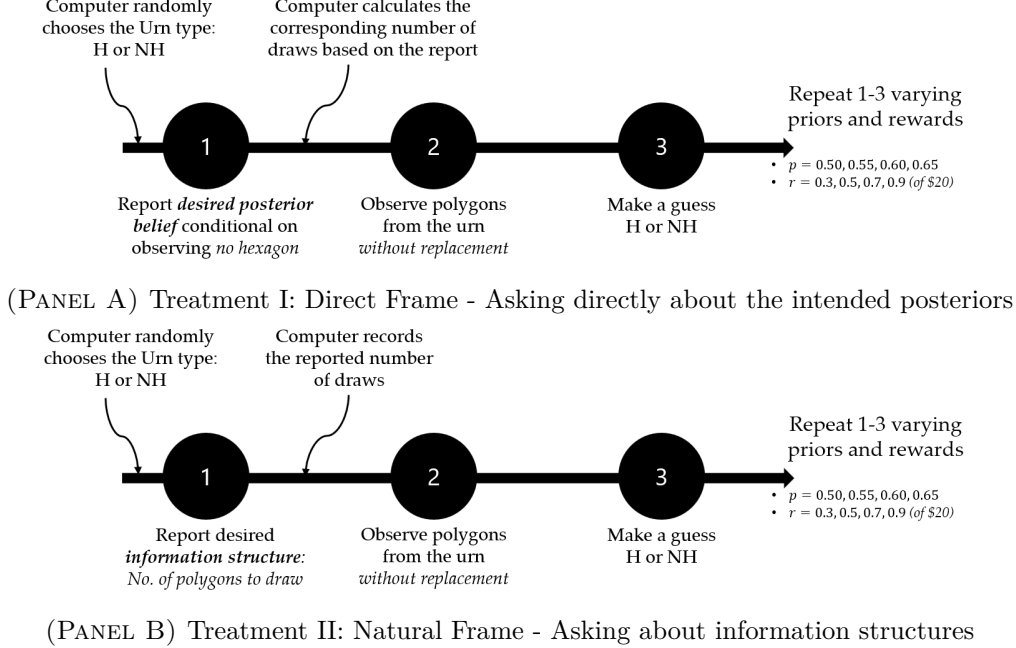


FIGURE 5. Timing of the Experiments - Section 1

3.2. Experiment 2: Tracking the Relevant Model

Without directly manipulating the representation of the information choice itself, the underlying cost class could instead be identified through its behavioral restrictions. For UPS costs, one such route builds on sequential optimization. [Bloedel and Zhong \(2025\)](#) provide a complementary optimization foundation. However, their analysis indicates that this criterion does not provide a symmetric way to compare the UPS and PI classes: commonly used PI specifications, such as likelihood-ratio costs, generally fail the criterion, although prior-invariance alone does not rule it out.

We therefore perform our experimental comparison by endogenizing the choice of participants over whether to make an incentivized information decision in the Posterior-based or the Experiment-based representation, not by enabling sequential optimization or optimal stopping. This endogenous choice provides auxiliary evidence about which representation participants regard as more relevant to their decision-making process and allows us to examine whether the representation-dependent patterns documented in the first part persist when the participant

selects the elicitation frame. At the same time, the design preserves a one-shot information-acquisition problem: participants attend to all components of a single signal and update their beliefs only once, limiting the additional complexity and fatigue that would arise from repeated updating and optimal-stopping decisions.⁸

Experiment 2 implements the same underlying decision problem as Experiment 1. We again vary the prior p and the reward r , but restrict the reward levels (r) to 50, 70, and 90 percent, excluding the 30 percent reward level used in Experiment 1. Because the 30 percent and 50 percent reward levels produced similar responses in Experiment 1, pooling them into a 50 percent level and reducing the number of reward levels allows us to shorten the experiment and limit fatigue from repeated decision making. Experiment 2 consists of 92 rounds divided into four parts. Two rounds are randomly chosen for payment at the end of each session.

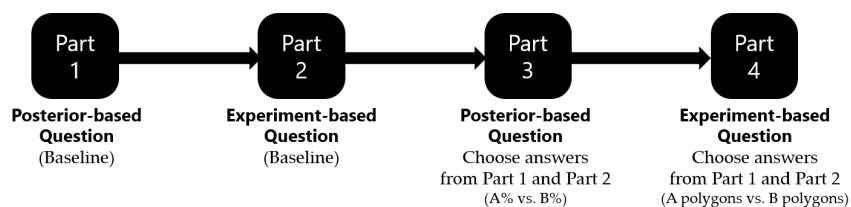


FIGURE 6. Timing of the Experiments - Section 2

In Parts 1 and 2, subjects face a protocol similar to that of Experiment 1. Part 1 uses the Posterior-based question, in which the decision variable is the desired accuracy level of inspection, whereas Part 2 uses the Sampling-based question, in which the decision variable is the sampling fraction. Parts 1 and 2 each consist of 29 rounds. In each round, subjects face a fixed prior p and specify their desired level of information for each of three urns associated with reward levels of 50, 70, and 90 percent. The interface the subject faces is shown in Figure 7. After the subject makes all three choices, the computer randomly selects one of the three urns and implements the information choice specified for that urn to determine the number of polygons drawn in that round. Each round therefore contains three contingent information choices, one for each reward level, although only one is randomly selected for implementation.

Parts 3 and 4 each consist of 17 rounds and use the same underlying decision environment. As in Parts 1 and 2, each round is characterized by a fixed prior p and three urns corresponding

⁸We hereby define our perspective of rational inattention by having a single signal and updating once. Even though the decision maker makes decisions sequentially, the behavior is captured aggregately ex-post in our perspective.

to the three reward levels. The choice set, however, is restricted to two information levels: the exact levels previously chosen by the same subject in Parts 1 and 2 for the corresponding combination of p and r . Thus, for each prior–reward pair, subjects choose between their own two previously elicited information choices. The interface, as Figure 8 shows, explicitly indicates whether each option originates from Part 1 or Part 2. Part 3 presents this choice using the Posterior-based representation, whereas Part 4 presents the same choice using the Experiment-based representation.

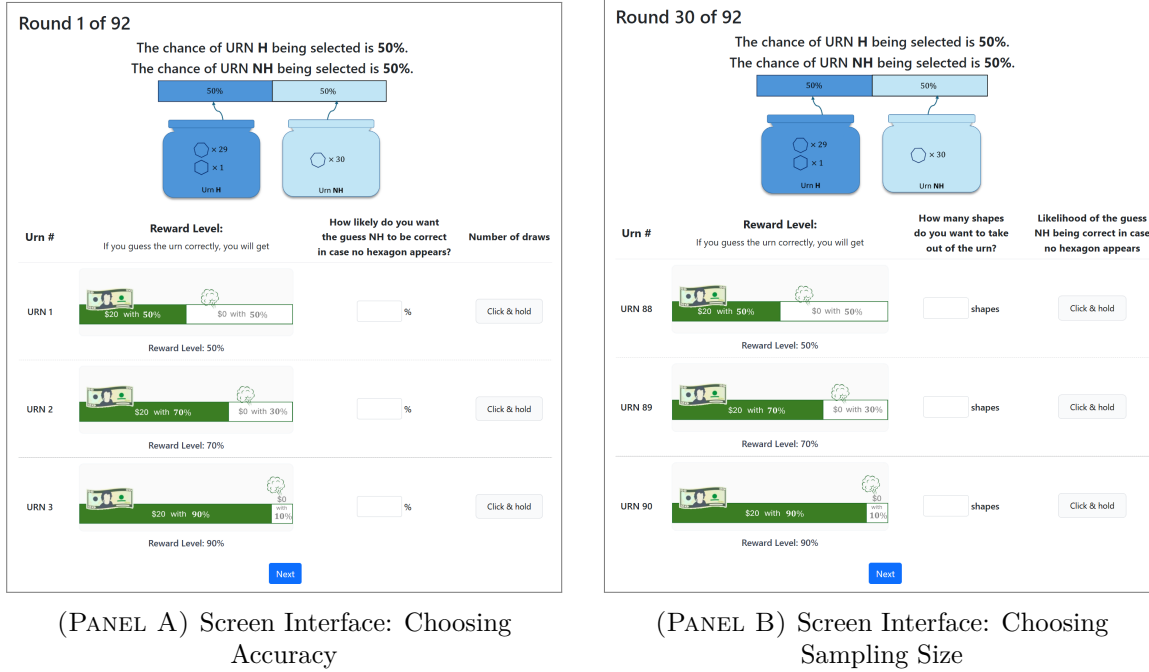
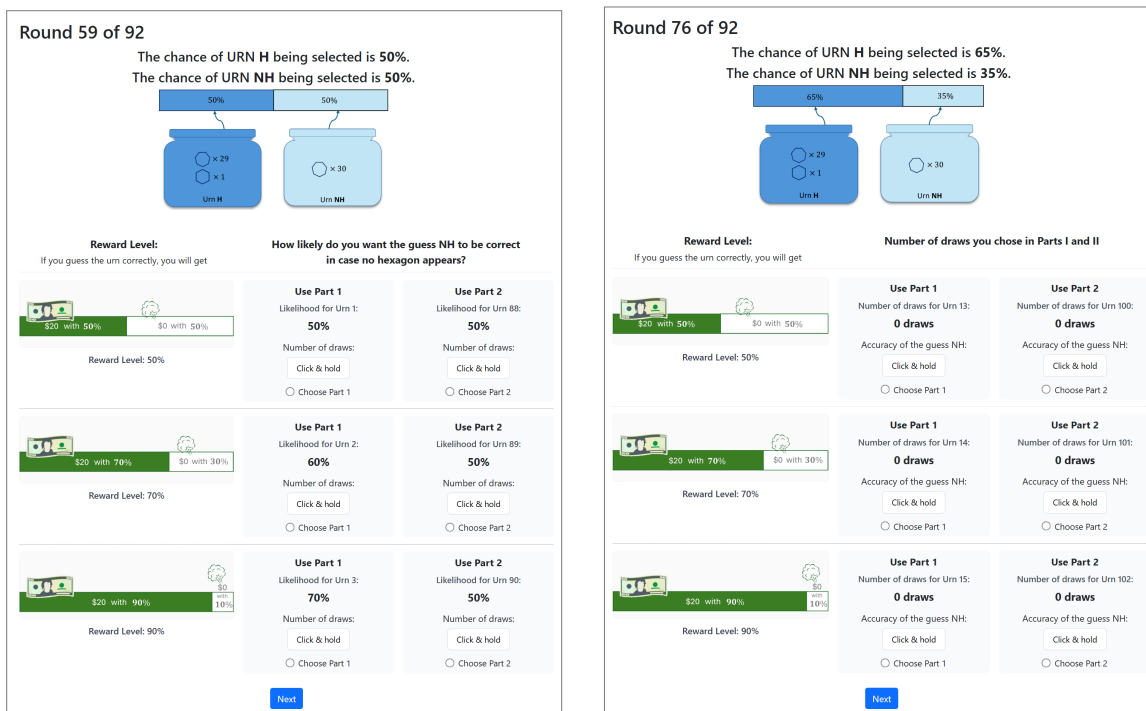


FIGURE 7. Example Screens of Part 1, 2

This design allows us to examine whether subjects prefer the information level previously chosen under one representation when the two previously selected levels are placed in direct comparison. Because the feasible alternatives in Parts 3 and 4 are determined entirely by the subject’s own earlier choices, these parts provide a within-subject comparison while holding fixed the underlying prior (p), reward (r), and pair of feasible information levels.

Across all four parts, the interface also provides buttons that allow subjects to verify the mapping between the decision variables used in the two representations, similar to the slider provided in the “Sampling-based+” treatment of Experiment 1. These tools are intended to reduce the role of mistakes arising from non-Bayesian updating or from misunderstanding the correspondence between the two representations. To control for potential order effects, we also



(PANEL A) Screen Interface: With
Posterior-based Question

(PANEL B) Screen Interface: With
Sampling-based Question

FIGURE 8. Example Screens of Part 3, 4: Choosing between the previous answers

counterbalance the order of Parts 1 and 2 and, separately, the order of Parts 3 and 4 across sessions.

4. RESULTS

The experiment was conducted at the Ohio State University in the year 2025-2026. The subject pool consists of Ohio State University students recruited via ORSEE (Greiner (2015)). In the Posterior-based treatment, 50 subjects participated; 41 in the Sampling-based treatment, and 34 in the Sampling-based+ treatment. The average payment was \$27.94 USD, including a \$10 USD show-up fee. Each experimental session lasted from 30 to 120 minutes, depending on the individual. The total number of observations is 21,000.

4.1. Rationally Inattentive behaviors

We first check if the subjects' behaviors can be represented as rationally inattentive behaviors by testing two conditions: no improving attention cycles (NIAC) and no improving action

switches (NIAS) (Caplin and Dean, 2015). The advantage of our simple design is that the NIAS condition is easily satisfied by most subjects – if a subject guesses the urn as H whenever they see a hexagon, then we can see that this subject satisfies the NIAS condition. To test NIAC, we define an individual as responsive to the reward if an individual seeks more informative information when the reward level is higher, whether informativeness is measured by the intended random posterior or by the information structure itself. By examining the subjects' responses at both aggregate and individual levels, we find that most subjects are classified as rationally inattentive. Subjects adjust their attention in response to the changes in incentives, in line with the rational inattention model. Figure 9 depicts the responses in aggregate, and Table 1 summarizes the number of individuals who are rationally inattentive, strictly rationally inattentive, and not rationally inattentive.

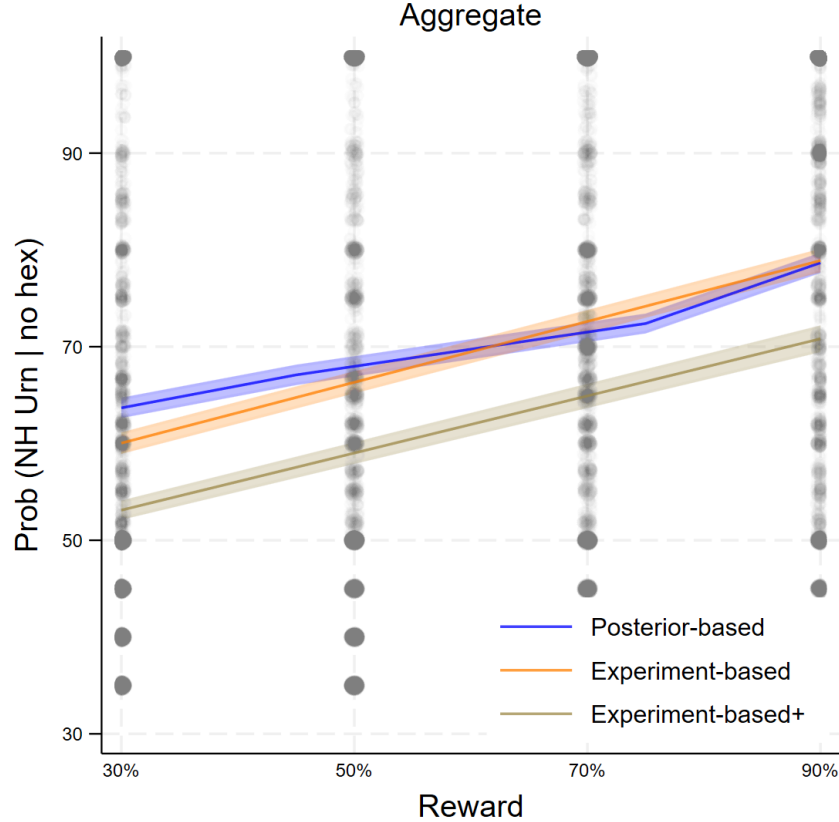


FIGURE 9. Responsiveness to Rewards (Aggregate)

$$\text{Prob}(\text{NH Urn}|\text{nh}) = \beta_0 + \beta_1 \text{reward} + \beta_2 \text{prior} + \epsilon$$

Treatment Group	$\beta_1 \geq 0$	$\beta_1 > 0$	$\beta_1 < 0$
Post-based	50	24	0
Samp-based	40	23	1
Samp-based+	34	19	0
All Sessions	124	66	1

TABLE 1. Responsiveness to rewards

From the data, we verify that most subjects, except only one observation in Sampling-based treatment, are rationally inattentive. The further analysis on model classifications are performed for these rationally inattentive subjects.

Result 1. *Most subjects are weakly responsive to the reward. At least half of the subjects are strictly responsive to the reward.*

4.2. Model Classifications and Model Validity in the lab

Second, we study *how* individuals bear information costs by testing responsiveness to variations in priors, which lets us close to identify the RI class that best allows behavior. This answers the question of *how* the information is revealed to be costly in the lab. *Both* patterns of behaviors that are classified as UPS and PI can be found in either frame, which shows the empirical validity of both models. However, the proportions of each model differ across the frames: more subjects are classified as UPS in the direct frame, whereas more are classified as PI in the natural frame. The aggregate level data in the figure 10 shows that the slopes of Natural and Natural+ groups is steeper than that of the Direct treatment group. The table 2 also determines these results by showing the number of individuals who may lie in each classification. This suggests that the question frame, which determines the decision primitive in the model, can affect the dominant class of behaviors.

Result 2. *Both UPS and PI are empirically valid, as well as the zero-cost model. The frames seemingly affect the dominant behavior.*

4.3. Consistency of Behaviors

Result 3. *The majority of subjects are consistent, though a significant portion switch mostly from UPS to PI. Bayesian belief mapping does not improve consistency. The two frames are not empirically equivalent, and the effect may be purely a matter of framing.*

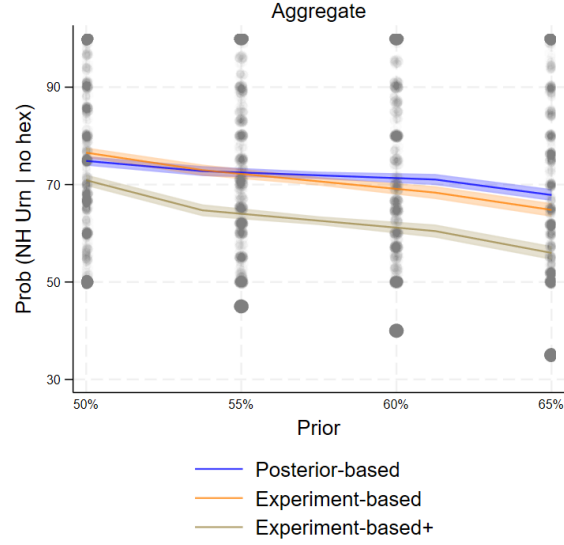


FIGURE 10. Responsiveness to Priors (Aggregate)

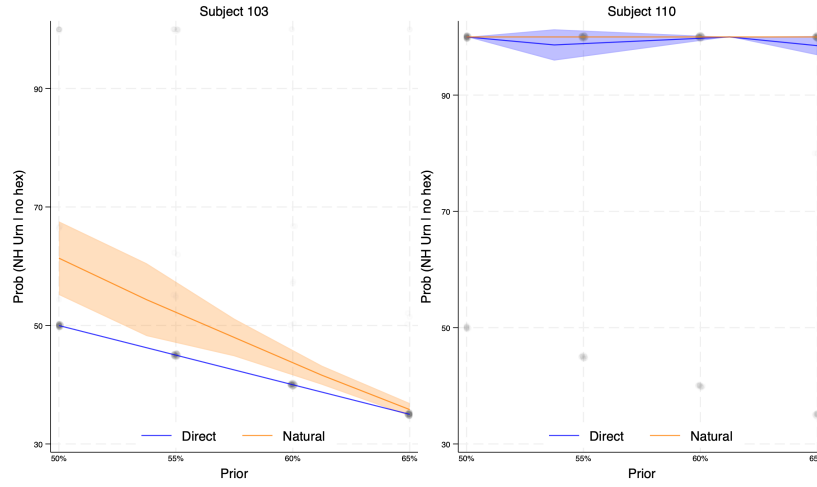


FIGURE 11. Examples of Individuals' Responsiveness to Priors

$$\text{Prob}(\text{NH}|\text{nh}) = \beta_0 + \beta_1 \text{reward} + \beta_2 \text{prior} + \epsilon$$

	$\beta_2 < 0$		$\beta_2 = 0$		Neither
	PI	UPS	Zero	High	
Pst-based	20	25	2	2	1
Natural	26	8	5	1	1 (not RI)
Natural+	25	7	0	2	0
Total	71	40	7	5	2

TABLE 2. Inattention Models (Individual)

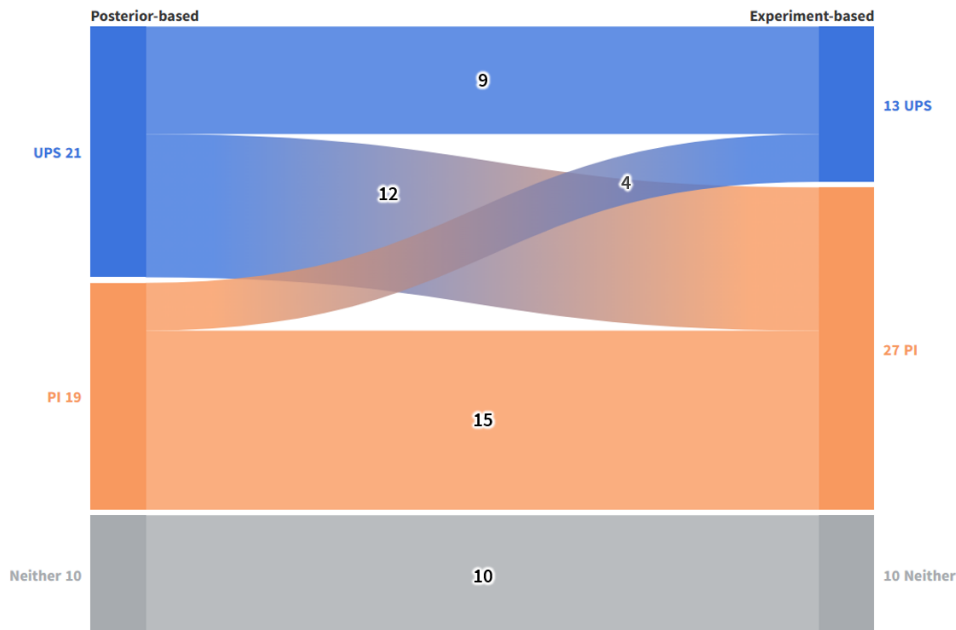


FIGURE 12. Behavioral Consistency

Lastly, we test the “practical” equivalence of subjects’ behaviors across the frames in the lab. That is, we investigate whether an individual’s behaviors are consistent across the frames. Although we find that many subjects have consistent classifications across the frames, a significant portion of people changed from UPS class in the Direct frame to PI class in the Natural frame, as shown in the table 3. The figure 13 following the table illustrates the decision patterns of the groups for the first four cells in the table 3.

	Natural UPS	Natural PI
Direct UPS	9 (zero cost: 2)	12
Direct PI	4	15
Consistent	24	
Switch	16	

TABLE 3. Behavioral Consistency (Individual)

This is surprising because the UPS and PI classes are (generically) theoretically disjoint, since their intersection admits only an individual with zero information costs. This finding calls into question the equivalence of subject behavior in the lab across the frames, though many previous empirical studies follow the theoretical equivalence between the random posteriors and information structures.

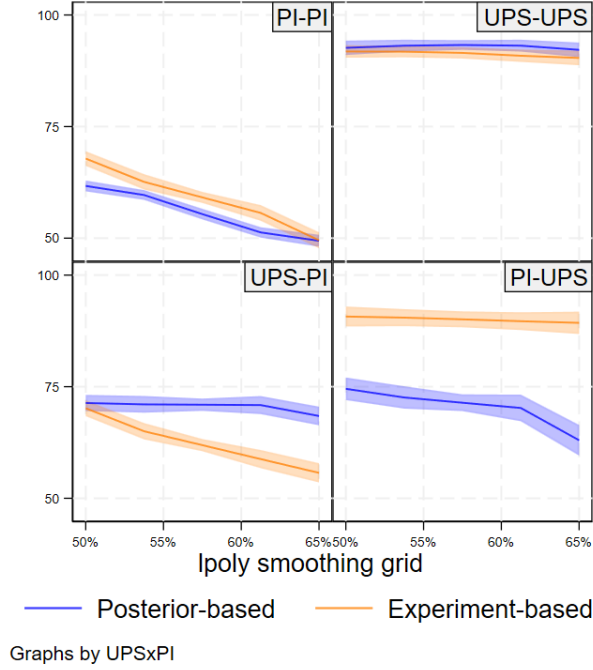


FIGURE 13. Behavioral Consistency (Aggregate)

In order to look closely into the ranges of this result and to eradicate the possibility of the non-Bayesian belief updating, i.e., a wrong mapping of the information structure and the posterior beliefs, we have another treatment, “Sampling-based+,” where the subjects are provided with the exact mapping of the number of balls they would take out and the accuracy percentage of the urn NH. However, the additional information regarding the belief mapping does not improve behavioral consistency, yielding results in the table 4 and the figure 14 similar to the earlier ones. Thus, the effect must be purely due to the different primitives under consideration, which is alarming from a theoretical perspective, as the intersection of the two classes of models allows only an individual with zero information costs.

	Natural+ UPS	Natural+ PI
Direct UPS	5	12
Direct PI	2	14
Consistent	19	
Switcher	14	

TABLE 4. Behavioral Consistency with Mapping information (Individual)

Result 4. *Frame affects accuracy regardless of consistency. The two frames are not empirically equivalent, and the effect may be purely a matter of framing.*

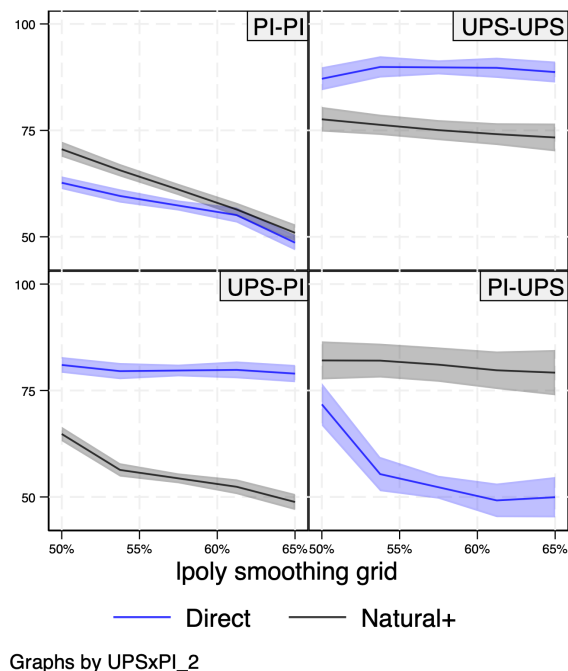


FIGURE 14. Behavioral Consistency with Mapping information (Aggregate)

To qualify this finding of non-equivalence of the two frames, we also see if one framework is more useful to make an accurate guess. We classify subjects into four groups based on the behaviors in each frame: UPS-UPS type, UPS-PI type, PI-UPS type, and PI-PI type. UPS-consistent(UPS-UPS) type subjects achieve high accuracy in both frames, regardless of whether an individual's classification is stable across frames.

This invites further work on *why* particular frames induce UPS versus PI behavior. As shown in Figures 15 and 16, inducing UPS-type behaviors yields higher correct rates, and this result closely follows the pattern of the chosen posterior beliefs. The more information that one desires to observe, the higher the correct guesses, which could be induced by suggesting one frame. We may argue that inducing UPS-type behaviors could be beneficial for an individual, suggesting a nudging implication toward UPS-type behaviors through different frames for considering different decision primitives.

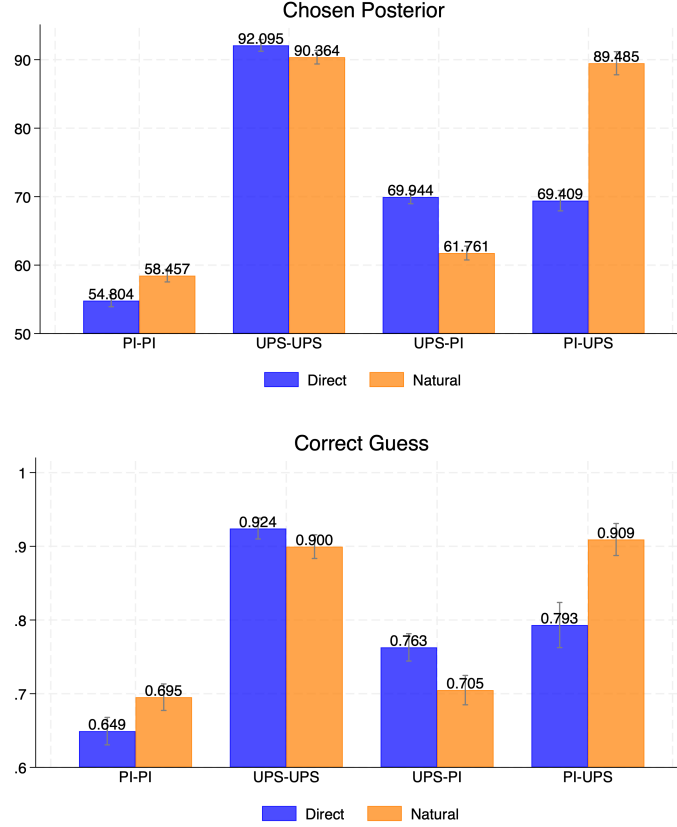


FIGURE 15. Correct Guesses by groups

4.4. Individual Model Relevance

5. CONCLUSION

This paper examines whether the information-cost structure revealed in a rational-inattention problem is stable across theoretically equivalent representations of information acquisition. In our experimental environment, each sampling choice corresponds one-to-one with a posterior-accuracy choice for any fixed prior. We exploit this equivalence to present subjects with the same feasible information choices in two ways. In the Posterior-based Treatment, subjects directly select the posterior accuracy they wish to obtain following a non-revealing signal. In the Experiment-based Treatment, subjects select the corresponding sampling size. The underlying states, payoffs, and feasible information technologies are held fixed across treatments.

We evaluate behavior under each representation using Prior-Invariant and Uniformly Posterior-Separable cost benchmarks. The experiment is not designed to map the treatments directly into the two cost classes. In particular, the Posterior-based Treatment is not defined as a test

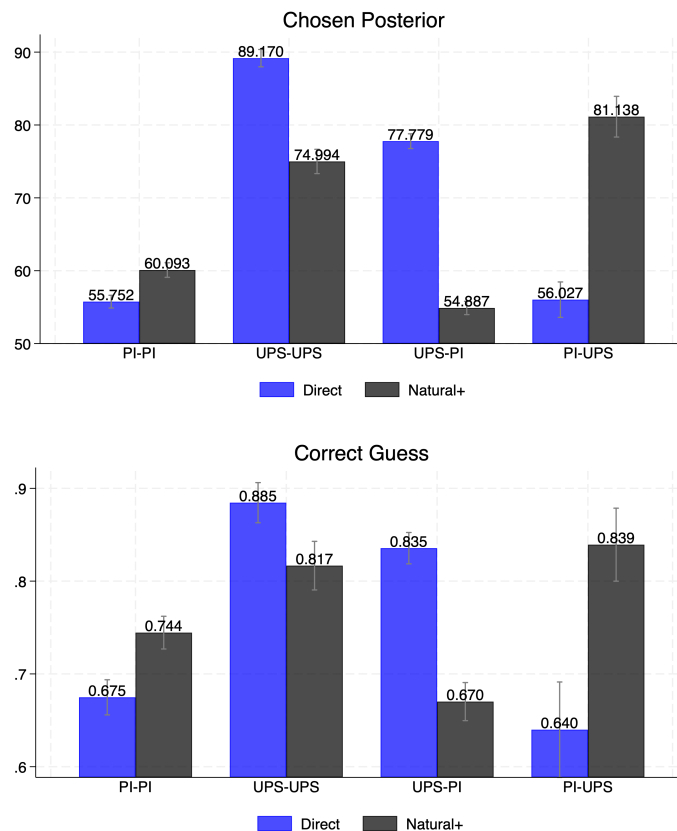


FIGURE 16. Correct Guesses by groups with Mapping information

of UPS costs, nor is the Experiment-based Treatment defined as a test of PI costs. Choices observed under either treatment may, in principle, be rationalized by either class. Our central question is instead whether changing the representation of the same information choice changes the pattern of information acquisition and the cost-class benchmark with which that pattern is most consistent.

Our results indicate that representation matters. Information acquisition differs systematically between the Posterior-based and Experiment-based Treatments even though the feasible information choices and their material consequences are unchanged. A substantial share of subjects exhibit patterns that are not fully consistent with either benchmark, and many subjects change the cost class with which their behavior is more consistent across treatments. At the aggregate level, the comparative statics of information acquisition also resemble different information-cost structures under the two representations. The observed differences therefore extend beyond a change in the units used to express a common information choice.

These findings should not be interpreted as a general establishment that posterior-based choices reveal UPS costs while experiment-based choices reveal PI costs. Rather, they qualify the interpretation of a revealed information cost as a representation-free structural primitive. The representation may affect which attributes of information acquisition become salient when the decision maker formulates the problem. The Posterior-based Treatment foregrounds the accuracy of the belief produced by information, whereas the Experiment-based Treatment foregrounds the extent of the investigation used to produce that belief.

The second section of the experiment provides complementary evidence by allowing subjects to choose the representation under which they make an incentivized information decision. This exercise reveals which representation the subject regards as more relevant for formulating the decision and allows us to examine whether the treatment-dependent patterns documented under externally assigned representations persist when the representation is endogenously selected. At the same time, the design preserves a one-shot information-acquisition problem and avoids the additional complexity associated with repeated belief updating and endogenous stopping decisions.

The results have implications for the empirical analysis of rational inattention. Existing revealed-preference and experimental approaches typically recover or classify information costs from behavior observed within a particular task and representation. The resulting cost is then often interpreted as a property of the decision maker and applied to other decision environments. Our findings identify an additional requirement for such an interpretation: the revealed cost structure must remain stable across equivalent representations of the information choice. Without this stability, a cost specification that organizes behavior under one elicitation method may fail to predict comparative statics under another, even when an analyst can translate exactly between the available information options.

This issue may be consequential in applications in which information costs determine equilibrium responses, welfare conclusions, or the effects of policy interventions. Theoretical work already demonstrates that the distinction between experiment-based and posterior-based costs can matter when beliefs and information structures are endogenous. Our findings suggest that the representation used to elicit information acquisition may also matter empirically. Estimates recovered from a posterior-based task therefore need not transport without qualification to an experiment-based decision environment, and the converse is also true.

Our experimental environment deliberately restricts attention to a binary state space and a one-dimensional family of information technologies. This restriction allows us to construct a transparent one-to-one mapping between sampling sizes and posterior accuracies and to hold the feasible information set fixed across treatments. It also limits the scope of what can be identified. The experiment evaluates whether behavior is consistent with the restrictions of PI and UPS costs on the experimental domain; it does not nonparametrically identify the information cost outside that domain. Future work may examine whether representation dependence persists with richer signal technologies, larger state spaces, sequential acquisition, and naturally occurring information tasks.

More broadly, the paper distinguishes theoretical equivalence from behavioral equivalence. For a fixed prior, an analyst may move without loss between an experiment and the posterior distribution it induces. A decision maker need not organize the two representations in the same way. Establishing representation invariance is therefore an important prerequisite for treating revealed information costs as consistent structural primitives of rationally inattentive behavior.

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Supplementary Materials

APPENDIX A INSTRUCTIONS FOR THE EXPERIMENTS

A.1 Instructions for Experiment 1 – Treatment 1: Posterior-based Question

Instruction of the Experiment

Welcome, and thank you for your participation in this Economics Experiment. This experiment will last for at most 2 hours. Please read the instructions carefully. You can earn a considerable amount of money depending on your decisions and chances. Your earnings will be paid out in cash after the end of the experiment.

Before we begin, we ask that you turn off your cell phones for the duration of this experiment. We also ask that you do not communicate with anyone for the duration of this experiment and only use the software provided to you on your devices. Failure to comply with these rules may result in dismissal from this experiment, and you will not be paid any earnings you may have otherwise received.

Overview

Our experiment consists of three sessions. We encourage you to participate in all three sessions. You are participating in the first session. This session consists of 168 rounds of decision-making.

In each round, your task is to guess whether the randomly selected urn is either *Hexagon (H)* or *No Hexagon (NH)*. To make a better guess, you may inspect this selected urn. After inspection, you finalize your guess and move to the next round.

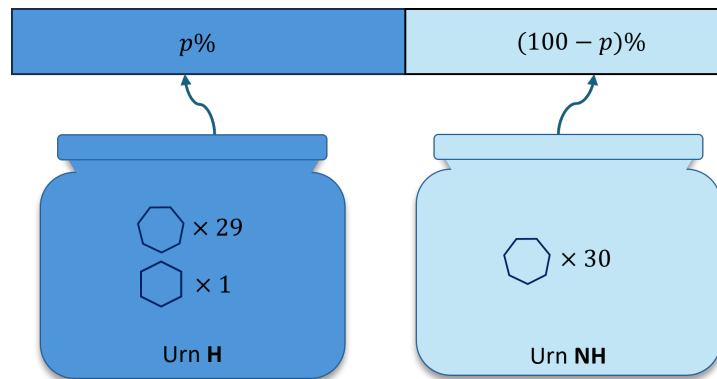
Guessing correctly gives you a chance to win a prize for your decisions in each round. The chance to win a prize will vary from round to round. Your final earnings will be based on your decisions from two randomly chosen rounds. Depending on your decisions, you can earn up to \$50 including \$10 of show-up fee. The details of the task follow.

Task

In each round, **your main task is to guess which urn has been selected, either *Hexagon (H)* or *No Hexagon (NH)*.**

Randomly Selected Urn

At the beginning of each round, the computer randomly selects an urn: H or NH . Each urn contains 30 polygons – there is **one hexagon (6-sided polygon) in urn H** and the remaining are septagons (7-sided polygon), and **all are septagons in urn NH** . For each round, there is a certain chance that the selected urn is H . The chance $p\%$ of H to be selected may vary for each round from 50% to 85% and is shown at the beginning of each round. With the remaining chance, the selected urn is NH .



The urn selected in one round does not affect the chances of the urn selected in other rounds.

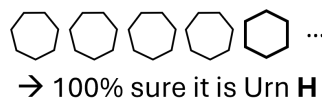
Inspecting the Selected Urn

To help you make a guess in a round, you have the opportunity to inspect the urn by taking out a number of polygons *without replacement*.

There are **two possible outcomes** from this inspection:

- 1) You may **observe a hexagon** during the inspection. In this case, you are certain that the selected urn is the urn *Hexagon (H)* since NH contains no hexagon. Therefore, the likelihood of the guess “ H ” being correct is 100% if you see a hexagon.

- If you see



- 2) You may **observe only septagons**. In this case, the polygons could have come from either urn, meaning there remains some uncertainty. That is, guessing “*No-Hexagon (NH)*” may be incorrect.

The process of this inspection is explained on the next page.

If you see
 ...    ...   (No hexagon)
 → Not sure whether it is
 Urn **H** or Urn **NH**
 → You believe $x\%$ is the chance that
 the urn type is **NH**
Your answer for the
 “Answering percentage”!

Process of Inspection

There are two steps to inspect the urn:

STEP 1: Answer the percentage

STEP 2: Observe the polygons taken out

The details of each step follow.

(STEP 1) Answering the percentage

The more accurate the inspection is, the more likely the guess **NH** is to be correct in the second case.

To determine the accuracy of your inspection, you will answer the following question:

“How likely do you want the guess **NH to be correct in case no hexagon appears?”**

You will answer the question in the form of a percentage(%). Notice that

- You will see more polygons if you choose a higher likelihood for the same initial chance of urn **NH**.
- You are not answering the number of polygons you want to take out. Your answer is a percentage.

(STEP 2) Observing the polygons taken out

We will then provide a number of polygons that are taken out for you to reach the likelihood you answer in step (1). The number of polygons provided to you is computed automatically, given your answer.⁹

Polygons are taken out randomly from the urn. Polygons that are taken out are not replaced, meaning any polygons taken from the urn will not be put back into the urn.

When taking out polygons, each draw takes 0.3 seconds to be taken out from the urn. The drawn polygon lasts 0.7 seconds on your screen, and the next draw will appear after that. You will continue to observe polygons up to the number based on your answer on the previous page.

Some Useful Scenarios:

⁹The formula for the number of polygons n is $30 \times (\text{likelihood}\% + p\%) - 1 \div (\text{likelihood}\% \times p\%)$ where $p\%$ is the chance that urn **H** is selected for the round.

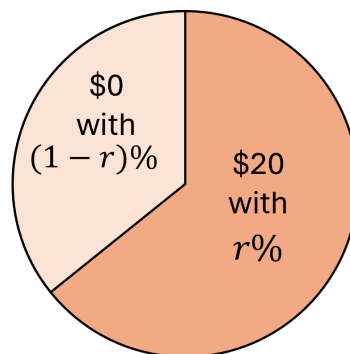
- If you answer with 100% in step (1), you will take out all 30 polygons in the urn. That is, if you want to inspect the urn surely and entirely, then you enter 100%.
- If you answer with $(100-p)\%$ —the chance the computer selects Urn NH—in step (1), you will not take out any polygons before guessing the urn. That is, if you do not want inspect the urn, you can enter the probability of NH (the probability shown in the lighter blue box).
- Entering the same probability in step (1) when the computer's chance to select NH is lower, like 35% versus 50%, takes out more polygons.

Making a Decision

After observing the sequence of polygons drawn from the urn, you will guess which of the urns, H or NH , is selected by answering the question, “**What is your guess about the urn?**”

Earnings

If you guess the urn correctly, you earn a chance $r\%$ to win \$20 presented in the **darker orange** slice of the pie. This $r\%$ chance of winning \$20 is given at the beginning of each round and may vary each round randomly from 30% to 90%. Note, there is some chance of $(100 - r)\%$ that guessing correctly does not win \$20. If your guess is incorrect, you do not have earnings for the round.



After completing all 168 rounds, the computer will randomly select two rounds for payment. The final payment from the experiment is determined as the sum of the earnings from the two selected rounds plus the show-up fee of the session, \$10.

Your Final Payment = (\$10 show-up fee) + (Earnings from two selected rounds)

Are there any questions so far?

Summary of the session

- The experiment has 168 rounds of decision-making in the first session.

- Each round involves guessing whether the selected urn is HEXAGON (H) or NO-HEXAGON (NH).
- Chance $p\%$ of selecting Urn H (shown in the darker blue box) may vary across rounds.
- Urn H contains one hexagon and 29 septagons; urn NH contains 30 septagons.
- You may inspect the selected urn to help make your guess.
- You choose how likely you want NH to be correct for the case you don't observe a hexagon.
- The number of polygons is automatically computed for you to reach your chosen likelihood.
- Polygons are drawn one by one, taking 0.3 seconds each, and displayed for 0.7 seconds.
- Correct guesses earn a chance of winning \$20 in a round, and two rounds are randomly selected for final payment, plus a \$10 show-up fee.
- Chance $r\%$ of winning \$20 (shown in the darker orange slice) varies from round to round.

A.2 Instructions for Experiment 1 – Treatment 2: Sampling-based Question

Instruction of the Experiment

Welcome, and thank you for your participation in this Economics Experiment. This experiment will last for at most 2 hours. Please read the instructions carefully. You can earn a considerable amount of money depending on your decisions and chances. Your earnings will be paid out in cash after the end of the experiment.

Before we begin, we ask that you turn off your cell phones for the duration of this experiment. We also ask that you do not communicate with anyone for the duration of this experiment and only use the software provided to you on your devices. Failure to comply with these rules may result in dismissal from this experiment, and you will not be paid any earnings you may have otherwise received.

Overview

Our experiment consists of three sessions. We encourage you to participate in all three sessions.

You are participating in the second session. This session consists of 168 rounds of decision-making.

In each round, your task is to guess whether the randomly selected urn is either *Hexagon (H)* or *No Hexagon (NH)*. To make a better guess, you may inspect this selected urn. After inspection, you finalize your guess and move to the next round.

Guessing correctly gives you a chance to win a prize for your decisions in each round. The chance to win a prize will vary from round to round. Your final earnings will be based on your decisions from two randomly chosen rounds. Depending on your decisions, you can earn up to \$50 including \$10 of show-up fee.

The details of the task follow.

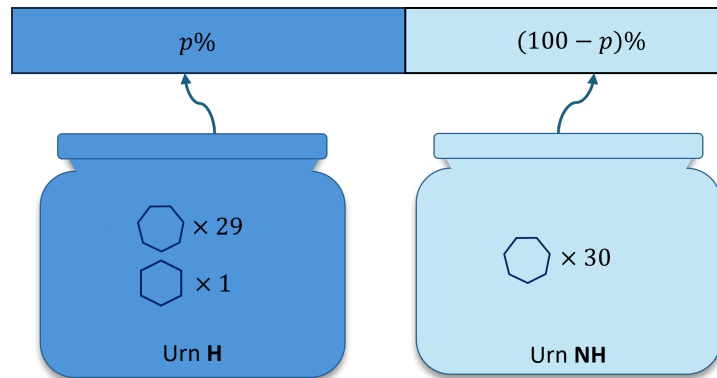
Task

In each round, **your main task is to guess which urn has been selected by the computer, either *Hexagon (H)* or *No Hexagon (NH)*.**

Randomly Selected Urn

At the beginning of each round, the computer randomly selects an urn: *H* or *NH*. Each urn contains 30 polygons – there is **one hexagon (6-sided polygon)** in urn *H* and the remaining are septagons (7-sided polygons), and **all are septagons in urn *NH***. For each round, there is a certain chance that

the selected urn is H . The chance $p\%$ of H to be selected may vary for each round from 50% to 85% and is shown at the beginning of each round. With the remaining chance, the selected urn is NH .



The urn selected in one round does not affect the chances of the urn selected in other rounds.

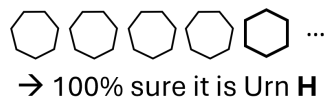
Inspecting the Selected Urn

To help you make a guess in a round, you have the opportunity to inspect the urn by taking out a number of polygons *without replacement*.

There are **two possible outcomes** from this inspection:

- 1) You may **observe a hexagon** during the inspection. In this case, you are certain that the selected urn is the urn *Hexagon (H)* since *NH* contains no hexagon. Therefore, the likelihood of the guess " H " being correct is 100% if you see a hexagon.

- If you see



- 2) You may **observe only septagons**. In this case, the polygons could have come from either urn, meaning there remains some uncertainty. That is, guessing "*No-Hexagon (NH)* " may be incorrect.

The process of this inspection is explained on the next page.

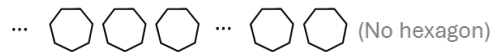
Process of Inspection

There are two steps to inspect the urn:

STEP 1: Answer the Number of draws

STEP 2: Observe the polygons taken out

If you see



→ Not sure whether it is
Urn **H** or Urn **NH**

→ You believe $x\%$ is the chance that
the urn type is NH

Your answer for the
“Answering percentage”!

The details of each step follow.

STEP 1: Answering the number of draws

The more polygons you draw, the more accurate your inspection is. To determine the accuracy of your inspection, you will answer the following question:

“How many polygons do you want to take out from the urn?”

You will answer the question in the form of a number (integer). Notice that

- Your answer must be an integer between 0 and 30.

(STEP 2) Observing the polygons taken out

We will then provide a number of polygons that are taken out for you. The number of polygons provided to you is exactly what you asked to take out in step (1).

Polygons are taken out randomly from the urn. Polygons that are taken out are not replaced, meaning any polygons taken from the urn will not be put back into the urn.

When taking out polygons, each draw takes 0.3 seconds to be taken out from the urn. The drawn polygon lasts 0.7 seconds on your screen, and the next draw will appear after that. You will continue to observe polygons up to the number based on your answer on the previous page.

Some Useful Scenarios:

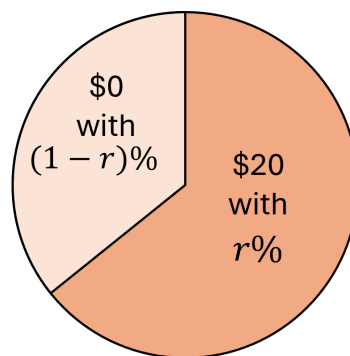
- If you answer 30 in step (1), you will take out all the polygons in the urn. That is, if you want to inspect the urn surely and entirely, then you enter 30.
- If you answer 0 in step (1), you will not take out any polygons before guessing the urn. That is, if you do not want to inspect the urn, you can enter 0.
If you take out more polygons,
- **Drawing less than a certain integer may not be possible since you are only allowed information that improves your chances of guessing correctly, but choosing no draws is always allowed.**

Making a Decision

After observing the sequence of polygons drawn from the urn, you will guess which of the urns, H or NH , is selected by answering the question, “**What is your guess about the urn?**”

Earnings

If you guess the urn correctly, you earn a chance $r\%$ to win \$20 presented in the **darker orange** slice of the pie. This $r\%$ chance of winning \$20 is given at the beginning of each round and may vary each round randomly from 30% to 90%. Note, there is some chance of $(100 - r)\%$ that guessing correctly does not win \$20. If your guess is incorrect, you do not have earnings for the round.



After completing all 168 rounds, the computer will randomly select two rounds for payment. The final payment from the experiment is determined as the sum of the earnings from the two selected rounds plus the show-up fee of the session, \$10.

Your Final Payment = (\$10 show-up fee) + (Earnings from two selected rounds)

Are there any questions so far?

Summary of the session

- The experiment has 168 rounds of decision-making in the first session.
- Each round involves guessing whether the selected urn is HEXAGON (H) or NO-HEXAGON (NH).
- The chance $p\%$ of the selected urn being H may vary across rounds.
- Urn H contains one hexagon and 29 septagons; urn NH contains 30 septagons.
- You can take out polygons from the urn to help make your guess.
- You specify the number of polygons to take out.
- Polygons are drawn one by one, taking 0.3 seconds each, and displayed for 0.7 seconds.

- Correct guesses earn a chance of winning \$20 in a round, and two rounds are randomly selected for final payment, plus a \$10 show-up fee.
- Chance $r\%$ of winning \$20 varies from round to round.

A.3 Instructions for Experiment 2

We conducted Experiment 2 in two orders. We present instructions for only one order, with Parts 1 and 3 as the “Posterior-based frame” and Parts 2 and 4 as the “Sampling-based frame.”

Instruction of the Experiment

Welcome, and thank you for your participation in this Economics Experiment.

This experiment will last for at most 2 hours. Please read the instructions carefully.

You can earn a considerable amount of money depending on your decisions and chances. Your earnings will be paid out in cash after the end of the experiment.

If you have any questions during the session, please do not hesitate to ask the experimenter.

Overview

This experiment consists of **four parts**. Each part contains either 29 or 17 rounds of decision-making, for a total of **92 rounds**.

In each round:

1. Your task is to guess whether the randomly selected urn is either *Hexagon (H)* or *No Hexagon (NH)*,
2. You may inspect this selected urn to make a better guess, and
3. You finalize your guess after inspecting the urn.
4. You have a short break in between rounds

Guessing correctly gives you a chance to win a prize for your decisions in each round. The chance to win a prize will vary from round to round.

Your final earnings will be based on your decisions from two randomly chosen urns. Depending on your decisions, you can earn up to \$50, including \$10 of a show-up fee.

Part I Instructions

This is the instruction for part I. The instructions for the other parts will appear before each part.

I. (Direct first) Task

In each round, **your main task is to guess which urn type has been selected by the computer, either *Hexagon (H)* or *No Hexagon (NH)*.**

Below is an example of a screen that you will see at the beginning of each round. We will go over each part of the screen one by one.

Round 1 of 92

The chance of URN H being selected is 50%.
The chance of URN NH being selected is 50%.

50%

50%

Urn H

Urn NH

Urn #	Reward Level: If you guess the urn correctly, you will get	How likely do you want the guess NH to be correct in case no hexagon appears?	Number of draws
URN 1	Reward Level: 50%	%	Click & hold
URN 2	Reward Level: 70%	%	Click & hold
URN 3	Reward Level: 90%	%	Click & hold

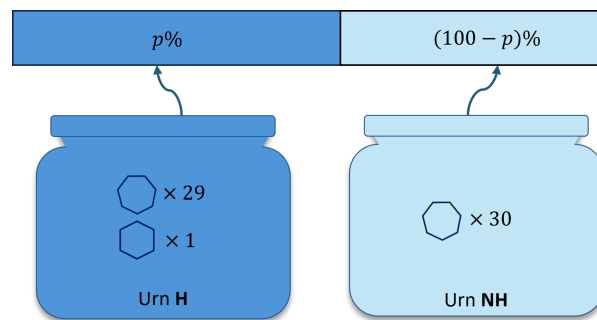
Next

II. Randomly Selected Urn

1. In each round, there are three urns corresponding to three reward levels.
2. Each of the urns contains 30 polygons.
3. There are two types of urns: **Urn H** and **Urn NH**
4. In **Urn H**, one is hexagon (6-sided shapes) and the remaining are septagons (7-sided shapes).
5. In **Urn NH**, all 30 shapes are septagons
6. How does the computer randomly select the urn types?

At the start of the round, we tell you the chance $p\%$ that the computer selects Urn H.

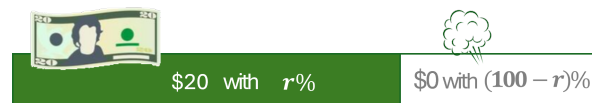
With the remaining chance $(100 - p)\%$ computer selects Urn NH.



7. The chance $p\%$ may vary for each round from 50%-65% and is shown in the figure.
8. The type of urn the computer selects for one urn does not affect the chances of the type selected for another urn.

III. Earning from an urn

1. There are three urns with different reward levels for each round.
2. The pie chart shows the reward level given by the chance $r\%$ of winning \$20 dollars.



3. Guessing the type of the urn correctly earns you a chance $r\%$ to win \$20 presented in the **green part** of the bar.
4. There is some chance of $(100 - r)\%$ that guessing correctly does not win \$20.
5. If your guess is incorrect, you do not have earnings for the round.

IV. (Direct first) Answering the percentage

1. For each urn, we will ask you the following question:

How likely do you want
the guess NH be correct,
in case no hexagon appears?

%

2. Your answer is the intended accuracy of your inspection of the urn.
3. It determines how many shapes you will take out from the urn without replacement.
 - To give you some idea of the number of shapes, we provide a calculated number in the column "*Number of draws*."

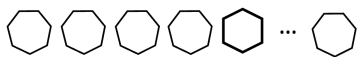
Number of draws

Click & hold

You can check the computed number while you click and hold the button in the column for the accuracy (%) you entered.

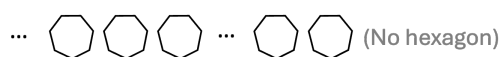
4. There are **two possible outcomes** from this inspection:

A) If you see a hexagon



→ You are 100% certain it is **Urn H**

B) If you only see septagons,



→ Not certain whether it is Urn **H** or **NH**

→ You are x% certain it is **Urn NH**

→ x% is your answer from the question
"How likely do you want ..."

5. After inspecting the urn, you will guess the type of the urn, H or NH, in the screen as below:

V. (Direct First) Inspecting the Selected Urn

1. One of the three urns on the previous page is randomly selected for the round's inspection.
2. You will inspect the urn based on your decision for the chosen urn.

Guess the type of the URN

You guess that the selected Urn type for this round is

☐ **NH** (No Hexagon) ☐ **N** (Hexagon)

Next

3. When taking out shapes, each draw takes 0.3 seconds to be taken out from the urn.
4. The shape will only last for 0.7 seconds on your screen.
5. New shapes will continue to be taken out until you reach the certainty $x\%$ based on your answer on the previous page.

Some Useful Scenarios:

1. If you want to inspect the urn surely and entirely, then you enter 100%.
2. If you do not want to inspect the urn, you can enter the probability of NH (the probability shown in the lighter blue box).
3. For the same level of desired accuracy, you will observe more shapes for rounds when the computer has a higher chance $p\%$ of selecting Urn H.

VI. Earnings

1. After completing all 92 rounds, two of the urns you have inspected are randomly chosen for payment.
2. The final payment from the experiment is determined as the sum of the earnings from the two selected rounds plus the \$10 show-up fee of the session.

Your Final Payment = (\$10 show-up fee) + (Earnings from two selected rounds)

Are there any questions so far?

(Direct first) Summary of the Part 1

- The experiment has 88 rounds of decision-making in the first session.
- Each round involves guessing whether the selected urn is HEXAGON (H) or NO-HEXAGON (NH).
- Chance $p\%$ of selecting Urn H (shown in the darker blue box) may vary across rounds.
- Urn H contains one hexagon and 29 septagons; urn NH contains 30 septagons.
- You may inspect the selected urn to help make your guess.
- You choose how likely you want NH to be correct for the case you don't observe a hexagon.

- The number of polygons is automatically computed for you to reach your chosen likelihood.
- Polygons are drawn one by one, taking 0.3 seconds each, and displayed for 0.7 seconds.
- Correct guesses earn a chance of winning \$20 in a round, and two rounds are randomly selected for final payment, plus a \$10 show-up fee.
- Chance $r\%$ of winning \$20 (shown in the darker orange slice) varies from round to round.
- You have a short break in between rounds.

(Direct first) Part 2 Instructions

This is the instruction for Part 2.

I. Task

In each round, **your main task is to guess which urn type has been selected by the computer, either *Hexagon (H)* or *No Hexagon (NH)*.**

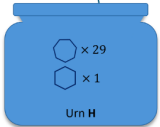
Below is an example of a screen that you will see at the beginning of each round. We will go over each part of the screen one by one.

Round 30 of 92

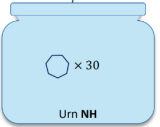
The chance of URN H being selected is 50%.
The chance of URN NH being selected is 50%.

50%

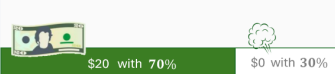
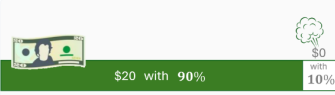
50%



Urn H



Urn NH

Urn #	Reward Level: If you guess the urn correctly, you will get	How many shapes do you want to take out of the urn?	Likelihood of the guess NH being correct in case no hexagon appears
URN 88	 Reward Level: 50%	shapes	Click & hold
URN 89	 Reward Level: 70%	shapes	Click & hold
URN 90	 Reward Level: 90%	shapes	Click & hold

Next

II. (Natural first) Answering the integer

1. For each urn, we will ask you the following question:

How many polygons do you want to take out from the urn?

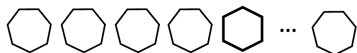
2. Your answer determines how many shapes you will take out from the urn without replacement.
 - To give you some idea of the accuracy of your inspection, we provide a calculated number in the last column.

Likelihood of the guess NH to
be correct in case no hexagon
appears

Press & hold

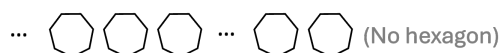
- You can check the computed accuracy of guessing NH (%) while you click and hold the button in the column for the number of draws you entered.
3. There are **two possible outcomes** from this inspection:

A) If you see a hexagon



→ You are 100% certain it is **Urn H**

B) If you only see septagons,



→ Not certain whether it is Urn **H** or **NH**

→ More septagons means
you are more certain it is **Urn NH**

4. After inspecting the urn, you will guess the type of the urn on the screen as below:

Guess the type of the URN

You guess that the selected Urn type for this round is

☐ **NH** (No Hexagon) ☐ **N** (Hexagon)

Next

III. (Natural First) Inspecting the Selected Urn

1. One of the three urns on the previous page is randomly selected for the round's inspection.
2. You will inspect the urn based on your decision for the chosen urn.
3. When taking out shapes, each draw takes 0.3 seconds to be taken out from the urn.
4. The shape will only last for 0.7 seconds on your screen.
5. New shapes will continue to be taken out until you reach the certainty $x\%$ based on your answer on the previous page.

Some Useful Scenarios:

1. If you want to inspect the urn surely and entirely, then you enter 30.
2. If you do not want to inspect the urn, you can enter 0.
3. You are only allowed to draw a number of shapes that improves your chances of guessing correctly, so some low numbers are not allowed.
4. But choosing to draw 0 is always allowed.

Are there any questions so far?

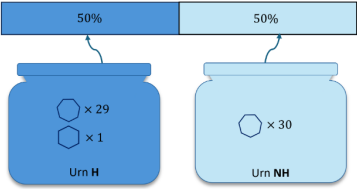
(Direct first) Part 3 Instructions

This is the instruction for Part 3.

Below is an example of a screen that you will see at the beginning of each round.

Round 59 of 92

The chance of URN H being selected is 50%.
The chance of URN NH being selected is 50%.



Reward Level:
If you guess the urn correctly, you will get

 \$20 with 50%	 \$0 with 50%
Reward Level: 50%	

How likely do you want the guess NH to be correct in case no hexagon appears?

Use Part 1	Use Part 2
Likelihood for Urn 1: 50%	Likelihood for Urn 88: 50%
Number of draws: Click & hold	Number of draws: Click & hold
☐ Choose Part 1	☐ Choose Part 2

Use Part 1	Use Part 2
Likelihood for Urn 2: 60%	Likelihood for Urn 89: 50%
Number of draws: Click & hold	Number of draws: Click & hold
☐ Choose Part 1	☐ Choose Part 2

Use Part 1	Use Part 2
Likelihood for Urn 3: 70%	Likelihood for Urn 90: 50%
Number of draws: Click & hold	Number of draws: Click & hold
☐ Choose Part 1	☐ Choose Part 2

Next

Task in Part 3

1. Like in Part 1, we ask you to answer “How likely do you want the guess NH to be correct, in the case no hexagon appears?”
2. Your options are now between the answers for the two previous urns in Part 1 and Part 2, respectively.

3. For each reward level, you may choose either to **Use Part I** answer or to **Use Part II** answer by using the radio button.
4. One of the three urns on the page is randomly selected for the round's inspection.
5. You will inspect the urn based on your decision for the chosen urn.
6. After inspecting the urn, you will guess the type of the urn, H or NH, in the screen as below:

Guess the type of the URN

You guess that the selected Urn type for this round is

☐ **NH** (No Hexagon) ☐ **N** (Hexagon)

[Next](#)

Are there any questions so far?

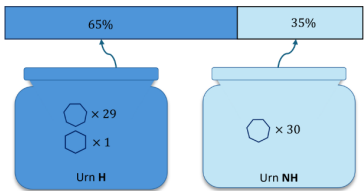
(Direct first) Part 4 Instructions

This is the instruction for Part 4.

Below is an example of a screen that you will see at the beginning of each round.

Round 76 of 92

The chance of URN H being selected is 65%.
The chance of URN NH being selected is 35%.



Reward Level:
If you guess the urn correctly, you will get

Reward Level	Use Part 1	Use Part 2
 Reward Level: 50%	Use Part 1 Number of draws for Urn 13: 0 draws Accuracy of the guess NH: Click & hold ☐ Choose Part 1	Use Part 2 Number of draws for Urn 100: 0 draws Accuracy of the guess NH: Click & hold ☐ Choose Part 2
 Reward Level: 70%	Use Part 1 Number of draws for Urn 14: 0 draws Accuracy of the guess NH: Click & hold ☐ Choose Part 1	Use Part 2 Number of draws for Urn 101: 0 draws Accuracy of the guess NH: Click & hold ☐ Choose Part 2
 Reward Level: 90%	Use Part 1 Number of draws for Urn 15: 0 draws Accuracy of the guess NH: Click & hold ☐ Choose Part 1	Use Part 2 Number of draws for Urn 102: 0 draws Accuracy of the guess NH: Click & hold ☐ Choose Part 2

Task in Part 4

1. Like in Part 2, we ask you to answer “*How many shapes do you want to take out from the urn?*”
2. Your options are now between the answers for the two previous urns in Part 1 and Part 2, respectively.
3. For each reward level, you may choose either to **Use Part I** answer or to **Use Part II** answer by using the radio button.

4. One of the three urns on this page is randomly selected for the round's inspection.
5. You will inspect the urn based on your decision for the chosen urn.
6. After inspecting the urn, you will guess the type of the urn, H or NH, in the screen as below:

Guess the type of the URN

You guess that the selected Urn type for this round is

☐ **NH** (No Hexagon) ☐ **N** (Hexagon)

Next

Are there any questions so far?

APPENDIX B SUPPLEMENTAL THEORETICAL BACKGROUNDS

B.1 Representations of Information

We investigate how the different representations of information, in terms of the experiment versus the posterior distribution, can affect the information acquisition behavior.

Since the posterior belief $\gamma_0(0) = \frac{1-p}{1-px}$, rearranging this yields $px = 1 - \frac{1-p}{\gamma_0}$. (From now on, we will just write $\gamma_0 \equiv \gamma_0(0)$ for simplicity, although abusing some notations.) This equation connects the choice of information in terms of “posterior belief” and “experiment,” mapping one to the other in a one-to-one manner.

$$p x(\gamma_0) = 1 - \frac{1-p}{\gamma_0}.$$

We can see from the equation that the differentials of x and γ_0 have this relation:

$$(1) \quad p dx = \left(\frac{1-p}{(\gamma_0)^2} \right) d\gamma_0$$

This indicates the “exchange rate” between x and γ_0 when translating the small change in x to that in γ_0 .

$$\frac{dp}{dx} < \frac{1-x(p)}{p(1-p)} \leq \frac{1}{p^2}.$$

Now we interchangeably translate the optimal decision represented in one to the other, using the exchange rate above. Under the environment in the assumption, we can express the gross payoff G as

$$\begin{aligned} G(p, r, x) &= \sum_{\gamma \in \Gamma(\pi)} \pi(\gamma) \left[\max_{a \in A} \sum_{\omega \in \Omega} \gamma(\omega) u(a, \omega) \right], \\ &= r(px + 1 - p) \end{aligned}$$

The gross payoff can be written in terms of the posterior γ_0 as

$$\begin{aligned} G(p, r, x(\gamma)) &= r(px + 1 - p) \\ &= r \left(1 - \frac{1-p}{\gamma_0} + 1 - p \right) \\ &= r \left(2 - p - \frac{1-p}{\gamma_0} \right) = G(p, r, \gamma_0) \end{aligned}$$

The decision maker chooses the optimal informativeness γ_0 that solves $G(p, r, \gamma_0) - K(p, Q(\gamma_0))$. From the First-Order-Condition, we have $\frac{\partial}{\partial \gamma_0} G(p, r, \gamma_0) = \frac{\partial}{\partial \gamma_0} K(p, \gamma_0)$, which gives

$$(2) \quad \begin{aligned} \frac{\partial}{\partial \gamma_0} K(p, \gamma_0) &= \frac{r(1-p)}{(\gamma_0)^2}, \text{ or} \\ (\gamma_0)^2 \frac{\partial K(p, \gamma_0)}{\partial \gamma_0} &= r(1-p) \end{aligned}$$

The optimal γ_0 that satisfies the above equation can be translated to the optimal experiment x using the “exchange rate” above. To be specific,

$$\begin{aligned} \frac{\partial K(p, \gamma_0)}{\partial \gamma_0} &= \frac{r(1-p)}{(\gamma_0)^2} \\ &= r \left(p \frac{dx}{d\gamma_0} \right) \\ \Leftrightarrow \frac{\partial K(p, \gamma_0)}{\partial \gamma_0} \frac{d\gamma_0}{dx} &= rp \frac{dx}{d\gamma_0} \frac{d\gamma_0}{dx} \\ \frac{\partial K(p, \gamma_0)}{\partial x} &= rp \end{aligned}$$

by the change of variables. This shows that choosing the optimal γ_0 that satisfies the FOC in equation (2) is exactly equivalent to choosing the optimal x that satisfies the identical FOC in the form of the above equation.

We verify this by letting this decision-maker now choose the information in terms of x . Using the gross payoff $G(p, r, x) = r(px + 1 - p)$, the decision maker now chooses the optimal information x that maximizes $G(p, r, x) - K(p, x)$. Solving FOC gives the optimal x that satisfies

$$(3) \quad \begin{aligned} \frac{\partial}{\partial x} G(p, r, x) &= \frac{\partial}{\partial x} K(p, x) \\ \Leftrightarrow \quad rp &= \frac{\partial K(p, x)}{\partial x} \end{aligned}$$

which is the same result that we obtained above.

Theoretically, the decision maker is assumed to choose the optimal information level regardless of the different representations of information. If there is a difference in the information choice depending on how the information is represented, especially given the Bayes updating, then how should this difference be understood?

We assume that the decision maker is able to choose optimal information in either representation. However, suppose we observe that optimal information choice in each representation

is not equivalent. This implies that the optimization problem this decision maker faces under each information representation is different, and that the information choice that maximizes their utility under one representation does not do so anymore when represented in the other way. Then there will be an apparent preference between the two information choices since one choice provides more welfare under a specific representation.

Furthermore, if the decision maker perceives the decision problem differently under each representation, but this difference comes from the additional cost that one representation may incur ¹⁰, then we may observe that this decision maker has a strict preference between the representations themselves. In such a case, we can argue that the decision maker follows one way of representing information whenever they face the information acquisition problem. In other words, this decision maker may always think of the problem in terms of the posterior distribution or in terms of the experiment.

B.2 Proofs of Proposition 1

B.3 Proofs of Proposition 2

B.4 Examples of UPS and PI Cost Functions

Example: UPS and PI Cost Functions. — Suppose a decision maker behaves following the cost function,

$$K^{UPS}(p, \gamma) = k [h(1 - p) - \mathbb{E}_Q[h(\gamma)]],$$

where the constant $k > 0$, and $h(\gamma) = -\gamma \ln \gamma - (1 - \gamma) \ln(1 - \gamma)$ is a binary entropy. (We employ the natural logarithm, having the unit of information as “nats.” We use the convention $0 \log 0 = 0$.) This is a Shannon mutual-information cost, exemplifying a UPS class cost function. In particular, it can be written according to the Definition 1, using $T(\gamma) = k [\gamma \ln \gamma + (1 - \gamma) \ln(1 - \gamma)]$, as

$$K^{UPS}(p, Q) = \mathbb{E}_Q[T(\gamma)] - T(1 - p).$$

¹⁰The cost of information is modeled here in a reduced way, and we do not distinguish any mental cost of information from some physical cost such as processing time. Disentangling different forms of information cost is not a focus of this study but a venue that could be further investigated.

The same potential T applies at every prior. Then the decision maker's net utility in posterior coordinates is therefore

$$\begin{aligned} U^{UPS}(p, r, \gamma_0) &= G(p, r, \gamma_0) - K^{UPS}(p, \gamma_0) \\ &= r \left(2 - p - \frac{1-p}{\gamma_0} \right) - k \left[h(1-p) - \frac{1-p}{\gamma_0} h(\gamma_0) \right]. \end{aligned}$$

For an interior optimum, the first-order condition reduces to

$$r = -k \ln(1 - \gamma_0),$$

which yields

$$\gamma_0^{UPS,*} = 1 - \exp\left(-\frac{r}{k}\right).$$

Hence, whenever the solution lies in the feasible, action-relevant interior, $\frac{\partial}{\partial p} \gamma_0^{UPS,*} = 0$, showing that the optimal posterior is locally invariant to the prior.

For the PI class cost function, consider a logarithmic sampling cost

$$\begin{aligned} K^{PI}(p, \pi_x) &= C(\pi_x) = -k_p \ln(1 - x), \quad (k_p > 0) \\ \Leftrightarrow K^{PI}(p, \gamma_0) &= -k_p \ln \left[1 - \frac{1}{p} \left(1 - \frac{1-p}{\gamma_0} \right) \right] \end{aligned}$$

This cost function features increasing and strictly convex costs, with

$$C'(x) = \frac{k_p}{1-x} > 0, \quad C''(x) = \frac{k_p}{(1-x)^2} > 0.$$

With this functional form of the cost, we see that increasing the probability of detecting the revealing signal becomes progressively more costly, and the cost diverges as the experiment approaches perfect, exhaustive sampling, $x \rightarrow 1$. The decision maker therefore solves for the optimal experiment $x^{PI,*}$ that maximizes the utility

$$\begin{aligned} U^{PI}(p, r, x) &= G(p, r, x) - K^{PI}(p, \pi_x) \\ &= r(px + 1 - p) + k_p \ln(1 - x) \\ \Leftrightarrow U^{PI}(p, r, \gamma_0) &= G(p, r, \gamma_0) - K^{PI}(p, \gamma_0) \\ &= r \left(2 - p - \frac{1-p}{\gamma_0} \right) + k_p \ln \left[1 - \frac{1}{p} \left(1 - \frac{1-p}{\gamma_0} \right) \right]. \end{aligned}$$

The first-order condition gives

$$x^{PI,*}(p, r) = 1 - \frac{k_P}{rp}.$$

An interior solution requires $rp > k_P$. Moreover,

$$\begin{aligned} \frac{\partial x^{PI,*}(p, r)}{\partial p} &= \frac{k_P}{rp^2} > 0. \\ \gamma_0^{PI,*}(p, r) &= \frac{1-p}{1 - px^{PI,*}(p, r)} \\ &= \frac{1-p}{1-p + k_P/r}. \end{aligned}$$

The optimal posterior is strictly decreasing in the prior:

$$\frac{\partial \gamma_0^{PI,*}(p, r)}{\partial p} = -\frac{k_P/r}{(1-p + k_P/r)^2} < 0.$$

A higher prior probability of state 1 therefore increases the marginal benefit of sampling and leads to a more informative experiment.

Among the PI class cost functions, there are some cost functions that do not necessarily lead to $d\gamma_0/dp < 0$. (i.e., the converse of Proposition 2 does not generally hold.) Within the restricted family of experiments, consider the quadratic cost

$$\begin{aligned} K^{PI}(p, \pi_x) &= C(\pi_x) = \frac{k_P}{2} x^2, \quad k_P > 0 \\ \Leftrightarrow K^{PI}(p, \gamma_0) &= \frac{k_P}{2} \left[\frac{1}{p} \left(1 - \frac{1-p}{\gamma_0} \right) \right]^2 \end{aligned}$$

The quadratic specification captures a convex sampling cost: increasing the probability of detecting the revealing signal requires progressively greater attention effort. This cost is prior invariant because the cost assigned to a fixed experiment does not depend on the prior. The decision maker therefore solves for the optimal experiment $x^{PI,*}$ that maximizes the utility

$$\begin{aligned} U^{PI}(p, r, x) &= G(p, r, x) - K^{PI}(p, \pi_x) \\ &= r(px + 1 - p) - \frac{k_P}{2} x^2 \\ \Leftrightarrow U^{PI}(p, r, \gamma_0) &= G(p, r, \gamma_0) - K^{PI}(p, \gamma_0) \\ &= r \left(2 - p - \frac{1-p}{\gamma_0} \right) - \frac{k_P}{2} \left[\frac{1}{p} \left(1 - \frac{1-p}{\gamma_0} \right) \right]^2. \end{aligned}$$

Conditional on an interior optimum in the action-relevant region, the optimal experiment is

$$x^{PI,*}(p, r) = \frac{rp}{k_P}.$$

or equivalently,

$$\gamma_0^{PI,*}(p, r) = \frac{1-p}{1 - \frac{r}{k_P}p^2}.$$

The interior solution requires $0 < rp/k_P < 1$, together with the condition that signal 0 induces action 0. This posterior varies with the prior:

$$\frac{\partial \gamma_0^{PI,*}(p, r)}{\partial p} = \frac{\frac{r}{k_P}p(2-p) - 1}{\left(1 - \frac{r}{k_P}p^2\right)^2}$$

The quadratic PI example therefore holds fixed the cost of a given sampling experiment across priors, rather than the cost of attaining a given posterior. As the prior changes, the marginal benefit of sampling changes, leading the decision maker to adjust the optimal experiment and, consequently, the posterior belief it induces.